



FINAL PROJECT

TCP1121 – COMPUTER PROGRAMMING
TRIMESTER 1, 2022/2023 (Term 2215)

Lab Section: 1FL

Submission Date: 6 February 2022

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Table of Contents

1	Task Distribution.....	2
2	Assignment Evaluation Form.....	3
3	Introduction.....	4
4	Problem Statements.....	5
5	Objective.....	6
6	Detailed Description.....	8
7	Screenshot of the program.....	34
8	Reference.....	46
9	Appendix.....	47

Task Distribution

No.	Student ID	Student Name	Tasks Completed
1.	1211105012	YEO QUAN XIANG	Coding, flowchart, screenshot of the program
2.	1211309854	ALHEJAIRI, ABDULLA MOHAMED	Coding, pseudocode, flowchart
3.	1201102109	ASHVIN A/L RAGUNATH	Program scope
4.	1221303894	PHILIP YONG ZHI XIANG	Draft interface design, detailed description

Project Evaluation Form

Part 1 – Program and Documentation

Scale				
1 - Poor	2 - Average	3 - Satisfactory	4 - Good	5 – Excellent

Assessment Criteria	Marks
1. Content and Interface (5%) Content relevance, interface appropriateness, content organization and presentation	
2. Functionality (5%) The program must contain at least 3 of the functionalities (add, edit, delete, view and search)	
3. Object-oriented Implementation (5%) Apply the OOP concept and use classes to solve the problem	
4. Completeness (5%) Completeness of the C++ program and free from bugs and errors	
5. Documentation (10%)	
Sub-Total (30%) (A)	

Part 2 – Level of Understanding

Scale					
0 No present	1 – 2 Poor	3 – 4 Average	5 – 6 Satisfactory	7 – 8 Good	9 – 10 Excellent

STUDENT ID	STUDENT NAME	Level of understanding (10%) (B)	TOTAL MARKS (40%) (A+B)
1211105012	YEO QUAN XIANG		
1211309854	ALHEJAIRI, ABDULLA MOHAMED		
1201102109	ASHVIN A/L RAGUNATH		
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Introduction

The topic of the program is a COVID-19 vaccination system. The COVID-19 pandemic has resulted in a large-scale effort to administer vaccines to protect people from the virus. This program aims to provide a solution to manage the administration of these vaccines.

The main part of the program is a database system that can store information about vaccine recipients and their vaccination status. Healthcare workers can add, view, search, and edit this information using a user-friendly interface.

The key points of the program are efficient management of the vaccine administration process. Secure storage of vaccine recipient data. User-friendly interface for healthcare workers to add, view, search, and edit records. Real-time tracking of the number of vaccines administered and the number of people who have completed their vaccination series. Compliance with privacy and security regulations for medical data. The program could significantly impact the effort to control the spread of COVID-19 and ensure the safety of people receiving the vaccine.

Problem Statements

1. Managing a large volume of data on vaccine recipients and their vaccination status.
2. Tracking the administration of vaccines to ensure everyone receives the necessary number and is effective.
3. Streamlining the administration process to be more efficient.
4. Ensuring the security and privacy of sensitive medical information on vaccine recipients.
5. Monitoring the progress of vaccine administration with real-time data tracking.

Objectives

1. To develop a system that can efficiently manage large amounts of data on vaccine recipients and their vaccination status, meeting the requirements of healthcare facilities and ensuring data security and privacy.
2. To create a system that can accurately track the administration of vaccines to ensure everyone receives the necessary number of doses and to monitor the progress of the vaccine administration effort.
3. To design a user-friendly system that streamlines the vaccine administration process, making it more efficient for healthcare workers and providing real-time data tracking.
4. To develop a system that meets privacy and security regulations for medical data, ensuring the safety and protection of sensitive information on vaccine recipients.
5. To create a system that allows users to efficiently manage and track the administration of COVID-19 vaccines, providing real-time data on vaccine distribution and administration.

Program Scope

1. In main menu, user can choose to enter admin, user and exit.
2. In sign in admin menu, user have to key in the user password, if user or password is wrong, an error appears and after pressing any key it returns to main menu. If password is correct, 'Welcome Admin.' is shown.

For Admin

3. In admin menu, admin can add vaccine stock by selecting which center and number of vaccines to be added. Then, returns to admin menu.
4. Show Vaccine Stock: admin can see number of vaccines available and assigned to person.
5. Search Vaccination Record: allows admin to view the vaccination record.
6. Add Vaccination Record: enables to apply for vaccine. IC number has to be entered followed by date and vaccine manufacturer, if any error has been key in, 'y' should be entered to retype or press any other key to return to admin menu.
7. Delete Vaccination Record: allows to delete vaccination record, if a record that exist is found, the details of the vaccination shown, press 'y' is used as a confirmation to delete the record. If a wrong name or IC number is inserted, it shows 'Record Not Found' and returns to the admin menu after pressing any key.
8. Sign Out: to sign out of admin menu. Returns to main menu.

For User

10. User Registration: user have to enter the IC number, name, contact number and password. If all the details is correct, 'You are successfully registered:)' is shown and after pressing any key returns to User Sub-Menu. If any error in typed, user can press 'y' to retype or other key to return to menu.
11. Sign In: User have to key in the IC number, if an error is key in user can press 'y' to retype. When an IC number which is not registered is key in 'You are not registered, please register before logging in.' and 'Wrong Password' if a wrong password is key in. When an existing IC and correct password is inserted, it brings you to the User Menu.
12. Change password: User can change their password by inserting IC number, old password and new password.
13. Delete account: User can delete the unwanted accounts from register.
14. Apply vaccine: In user menu, user can apply vaccine by inserting their IC number, name, date to receive vaccination and manufacturer.
15. View Vaccination Record: user can check vaccination record when IC number and name is inserted.

16. Delete Vaccination Record: user must insert IC number and name followed by 'y' as a confirmation to delete.
17. Sign Out: To sign out of user menu. Returns to main menu.

Detailed Description

A COVID-19 vaccination system is implemented to administer the COVID-19 vaccine to the public in an organized and efficient manner. The goal of a COVID-19 vaccination system is to ensure that as many people as possible are vaccinated to reduce the spread of the virus and eventually reach herd immunity.

The system typically involves several key steps, including:

1. **Prioritization:** The first step in a COVID-19 vaccination system is determining who should be vaccinated first. This typically includes groups at high risk for severe illness, such as healthcare workers, elderly individuals, and those with underlying health conditions.
2. **Registration and Scheduling:** Once a person is eligible for the vaccine, they must register and schedule an appointment for their vaccine. This can be done through online portals, over the phone, or in person.
3. **Vaccination:** The person will then attend their appointment and receive the vaccine. Trained healthcare professionals, such as nurses or pharmacists, typically do this.
4. **Monitoring:** After receiving the vaccine, the person will be monitored for adverse reactions. They may also be asked to return for a second dose later to complete the vaccination series.
5. **Record Keeping:** Accurate record keeping is a vital component of any COVID-19 vaccination system. This includes keeping track of who had received the vaccine, when they received it, and any adverse reactions they may have experienced.

- **System design (use-case diagram, flow chart, pseudocode, etc)**

pseudocode

START

CREATE CLASS "covid_vaccination"

DECLARE VARIABLES name, password, contact, user_name, user_IC, IC, VacCenter, SumVacC2, SumVacCenter and add as protected variables.

CREATE STRUCTURE vac_data with variables uIC, uname, dose, manufacturer, and udate.

DECLARE COUNTER VARIABLES as public.

CREATE FUNCTION "title" to display the title of the program.

CREATE FUNCTION "retypeOpt" to retype the option.

CREATE FUNCTION "menu" to display the menu of the program.

CREATE FUNCTION "admin" for the admin to access the program.

CREATE FUNCTION "admin_password" for the admin to enter their password.

CREATE FUNCTION "user" for the user to access the program.

CREATE FUNCTION "user_password" for the user to enter their password.

CREATE FUNCTION "valid" to check if the user's IC exists.

```

CREATE FUNCTION "vac_center" to manage the vaccination center.
CREATE FUNCTION "add_vaccine_stock" to add the vaccine stock.
CREATE FUNCTION "display_vaccine_stock" to display the vaccine stock.
CREATE FUNCTION "vaccination_record" to record the vaccination.
CREATE FUNCTION "apply_vaccine" for the user to apply for a vaccine.
CREATE FUNCTION "view_vaccination" to view the vaccination records.
CREATE FUNCTION "delete_vaccination" to delete the vaccination records.
END CLASS

```

```

CREATE FUNCTION title()

```

```

CREATE FUNCTION retypeOpt()
Press any key to continue
CLEAR input
CLEAR cin buffer
CLEAR cout buffer

```

```

CREATE FUNCTION menu()
CLEAR screen
DECLARE integer option
Input option
call function by choosen num
case 1:
    BEGIN admin()
case 2:
    BEGIN user()
case 3:
    CLEAR screen
default:
    USE retypeOpt()
RETURN TO Main Menu

```

```

CREATE FUNCTION admin()
BEGIN admin_password()
CLEAR screen
DECLARE integer a_option
Input a_option
call function by choosen num
case 1:
    BEGIN add_vaccine_stock()
    GO TO Admin Menu
case 2:
    BEGIN display_vaccine_stock()
    GO TO Admin Menu

```

case 3:

BEGIN view_vaccination()

GO TO Admin Menu

case 4:

BEGIN vaccination_record()

GO TO Admin Menu

case 5:

BEGIN apply_vaccine()

GO TO Admin Menu

case 6:

BEGIN delete_vaccination()

GO TO Admin Menu

case 7:

GO TO Main Menu

default:

USE retypeOpt()

GO TO Admin Menu

CREATE FUNCTION admin_password()

CLEAR screen

DECLARE a_name

DECLARE a_password

Input a_name

Input a_password

If input a_name is "admin" and input a_password is "admin" then

PAUSE system

CLOSE system

Else other inputs "Wrong Name Or Password."

PAUSE system

CLOSE system

RETURN TO Main Menu

CREATE FUNCTION user()

BEGIN user_password()

CLEAR screen

DECLARE integer u_option

Input u_option

call function by choosen num

case 1:

BEGIN apply_vaccine()

GO TO User Menu

case 2:

BEGIN view_vaccine()

GO TO User Menu

case 3:

BEGIN delete_vaccination()

GO TO User Menu

case 5:

GO TO Main Menu

default:

USE retypeOpt()

GO TO User Menu

CREATE FUNCTION user_password()

CLEAR screen

DECLARE integer u_option

Store text for: fname, usern, userp

opening a file (fileo) to write contents into the file

opening a file (filei) to read contents from the file

Input u_option

if (u_option equal to 1)

CLEAR Screen

Input IC

if (IC.length() does not equal to 12)

Input choice

if choice is 'y' or 'Y'

Continue attempts

else

CLEAR input

CLEAR input buffer

GO TO User Menu

When counter equal 0 it is a valid IC

if counter is more than or equal to 3

Press any key to continue

user_password()

Ignore input

accept the user input in multiple lines until the delimiter character found(cin, name)

Input contact

Input password

fname = IC + ".txt"

Open fileo

Save name, password, contact in fileo

Press any key to continue

Close fileo

GO TO User Menu

else if u_option equal to 2

CLEAR screen

Store text for: u_IC, u_pass

```
Input u_IC
if (u_IC.length() does not equal to 12)
    Input choice
    if choice is 'y' or 'Y'
        Continue attempts
    else
        CLEAR input
        CLEAR input buffer
        GO TO User Menu
Input u_pass
fname = u_IC + ".txt"
Open filei
if filei is open() and/or filei fail
    Close filei
    Press any key to continue
    GO TO user_password
accept the user input in multiple lines until the delimiter character found(filei, usern)
accept the user input in multiple lines until the delimiter character found(filei, userp)
if (u_pass equals to userp)
    user_IC = u_IC
    user_name = usern
    Press any key to continue
else if (u_pass does not equal to userp)
    Input choice2
    if choice2 is 'y' or 'Y'
        Continue attempts
    else
        Press any key to continue
        CLEAR input
        CLEAR input buffer
        CLEAR output buffer
        GO TO user_password()
else if u_option equal to 3
    CLEAR screen
    Store text for: u_IC, u_pass
    Input u_IC
    if (u_IC.length does not equal to 12)
        Input choice
        if choice is 'y' or 'Y'
            Continue attempts
        else
            CLEAR input
            CLEAR input buffer
            GO TO User Menu
```

```

fname = u_IC + ".txt"
Open filei
if filei is open and/or filei fail
    Close filei
    Press any key to continue
    GO TO user_password
accept the user input in multiple lines until the delimiter character found(filei, usern)
accept the user input in multiple lines until the delimiter character found(filei, userp)
Close filei
Input u_pass
if (u_pass equals to userp)
    Store text for: newPassword
    Input newPassword
    opening a file by name fileo to write contents into the file
    Open fileo
    Save usern, newPassword in fileo
    Close fileo
    Press any key to continue
    CLEAR screen
    CLEAR input
    CLEAR input buffer
    CLEAR output buffer
    GO TO user_password()
else if (u_pass does not equal to userp)
    Press any key to continue
    CLEAR input
    CLEAR input buffer
    CLEAR output buffer
    GO TO user_password()
else if (u_option equals to 4)
    CLEAR screen
    Store text for: u_IC, u_pass
    Input u_IC
    if (u_IC.length() does not equal to 12)
        Input choice
        if (choice equal to 'y' or 'Y')
            GO TO user_password()
        else
            CLEAR input
            CLEAR input buffer
            GO TO User Menu
fname = u_IC + ".txt"
Open filei(fname.c_str())
if filei is open and/or filei is fail

```

```

    Close filei
    Press any key to continue
    GO TO user_password()
    accept the user input in multiple lines until the delimiter character found(filei, usern)
accept the user input in multiple lines until the delimiter character found(filei, userp)
Close filei
Input u_pass
if (u_pass equal to userp)
    int result = remove(fname.c_str())
    if (result equal to 0)
        Press any key to continue
        CLEAR input
        CLEAR input buffer
        CLEAR output buffer
        GO TO Main Menu
    else
        Press any key to continue
        CLEAR input
        CLEAR input buffer
        CLEAR output buffer
        user_password()
else if (u_option equal to 5)
    GO TO Main Menu
else
    USE retypeOpt()
    GO TO user_password()

```

```

CREATE FUNCTION valid(string str)
    Store text for: dir, user
    opening a file by name file to read contents from the file
    dir = str + ".txt"
    Open file to convert a string to an array of characters with a null character at the end.
    if file is open and/or file is fail
        return
    else
        counter++
        if (counter >= 3)
            Close file()
            return
    Input IC
    if (IC.length() does not equal to 12)
        Input choice
        if (choice equal to 'y' or 'Y')
            goto

```

```

    else
        CLEAR input
        CLEAR input buffer
        return
    GO TO valid(IC)

```

```

CREATE FUNCTION vac_center()
opening a file (vac_center.txt) to read contents from file
if file is fail
    Close file()
    Open file("vac_center.txt")
    file DEFAULT_VACCINE
    Close file()

```

```

CREATE FUNCTION add_vaccine_stock()
int option
opening a file to read contents from file1, file2, file3
CLEAR screen
    Input option
switch (option)
case 1:
    file1.open("vac_center.txt")
    Input add
    SumVacCenter = SumVacCenter + add
    file1 SumVacCenter
    file1.close()
    Press any key to continue
    goto
    break
case 2:
    break
default:
    retypeOpt()
    break

```

```

CREATE FUNCTION display_vaccine_stock()
CLEAR screen
vac_data data
open a file (ind) in read mode
opening a file to read contents from file
int a
int i = 0
Open file("vac_center.txt")
file a
read data from the file "Vaccination_Records.bin"

```



```

Set the input file's position to the beginning of the file
Read records from the input file until the end is reached
    Increment the record counter
Close file
int s = a - i
Close file()
Press any key to continue

```

```

CREATE FUNCTION vaccination_record()
    vac_data Data
    CLEAR screen
    open a file "Vaccination_Records.bin" in read mode
    opening a file by name Vaccination_Records.bin to read contents from the file
    if file is open "Failed to open file"
    Get the size of the file
    Allocate memory for the buffer
    Read contents of file into buffer
    Calculate the number of data structures in the buffer
    Determine maximum length for each column
    Display table header
    Display contents of data as a table
    for (int i = 0 i < dataCount ++i)
        vac_data d
        memcpy(&d, &buffer[i * sizeof(Data)], sizeof(Data))
    delete[] buffer
    Press any key to continue

```

```

CREATE FUNCTION apply_vaccine()
    vac_data data
    vac_data datas
    vac_data datac
    opening a file to read contents from file
    open files in read mode
    Opening files to read contents from the file
    int length
    int day, month, year
    int result
    int count = 0
    CLEAR screen
    CLEAR input buffer
    Input data.ulC
    if (strlen(data.ulC) does not equal to 12)
        Input choice
        if (choice equal to 'y' or 'Y')

```

```

        goto
    else
        CLEAR input
        CLEAR input buffer
        goto
    Open file("Vaccination_Records.bin")
    check IC exist before or not
    if (strcmp(data.uIC, datac.uIC) equal to 0)
        count++
    Close file()
    files.open("Vaccination_Records.bin")
    if (!files)
        data.dose = '1'
        check IC exist before or not
        data.dose = (char)(count + 49)
        strcpy(data.uname, datas.uname)
        goto
    else
        data.dose = '1'
    files.close()
    Input data.uname
    Input data.udate
    result = sscanf(data.udate, "%d/%d/%d", &day, &month, &year)
    if (result does not equal to 3)
        Input choice
        if (choice equal to 'y' or 'Y')
            goto
        else
            CLEAR input
            CLEAR input buffer
            goto
    Input data.manufacturer
    length = strlen(data.manufacturer)
    for (int i = 0; i < length; ++i)
        data.manufacturer[i] = tolower(data.manufacturer[i])
    if (strcmp(data.manufacturer, "pfizer-biontech") equal to 0)
        strcpy(data.manufacturer, "Pfizer-BioNTech")
    else if (strcmp(data.manufacturer, "astrazeneca") equal to 0)
        strcpy(data.manufacturer, "AstraZeneca")
    else if (strcmp(data.manufacturer, "sinovac") equal to 0)
        strcpy(data.manufacturer, "Sinovac")
    else
        Input choice
        if (choice equal to 'y' or 'Y')

```

```

        goto
    else
        CLEAR input
        CLEAR input buffer
        goto
    Input choice
    if (choice equal to 'y' or 'Y')
        Open file("Vaccination_Records.bin")
        Write to data to file
    else
        CLEAR input
        CLEAR input buffer
    Close file()
    Press any key to continue

```

```

CREATE FUNCTION view_vaccination()
    vac_data datac
    int count = 0
    opening a file to read contents from file
    CLEAR screen
    Open file("Vaccination_Records.bin")
    check file open
        goto
    Input bIC
    if (strlen(bIC) does not equal to 12)
        Input choice
        if (choice equal to 'y' or 'Y')
            goto
        else
            CLEAR input
            CLEAR input buffer
            goto
    Input bname
    read data from file
    if (strcmp(bIC, datac.uIC) equal to 0 && strcmp(bname, datac.uname) equal to 0)
        count++
        "Dose " datac.dose ": "
        "Date: "
        "Manufacturer: "
    if (count equal to 0)
    Close file()
    Press any key to continue

```

```

CREATE FUNCTION delete_vaccination()

```

```

vac_data datac
int count = 0
opening a file to read contents from file
CLEAR screen
Open file("Vaccination_Records.bin")
check file open
    goto
else
    goto
Input bIC
if (strlen(bIC) does not equal to 12)
    Input choice
    if (choice equal to 'y' or 'Y')
        goto
    else
        CLEAR input
        CLEAR input buffer
        goto
Input bname
read data in file
    if (strcmp(bIC, datac.uIC) equal to 0 && strcmp(bname, datac.uname) equal to 0)
        count++
        "Dose " datac.dose ": "
        "Date: "
        "Manufacturer: "
Close file()
if (count equal to 0)
    goto
Input choice
if (choice equal to 'y' or 'Y')
    open a file "Vaccination_Records.bin" in read mode
    opening a file by name Vaccination_Records.bin to read contents from the file
    if binaryFile has Error Opening Vaccination Record File
opening a file ("Vaccination_Records_temp.bin") to write contents into the file
if updated Binary File has Error Opening Vaccination Record Temp File
    Close binary File
read data from binary File
    if (strcmp(datac.uIC, bIC) does not equal to 0)
        write to Binary File to update data
    Close binary File
    Close updated Binary File
    remove("Vaccination_Records.bin")
    rename("Vaccination_Records_temp.bin", "Vaccination_Records.bin")
else

```

```

CLEAR input
CLEAR input buffer
goto
Press any key to continue

```

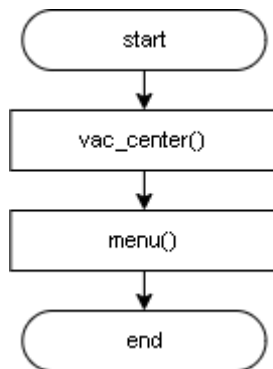
```

CREATE int main()
  covid_vaccination system
  system.vac_center()
  system.menu()

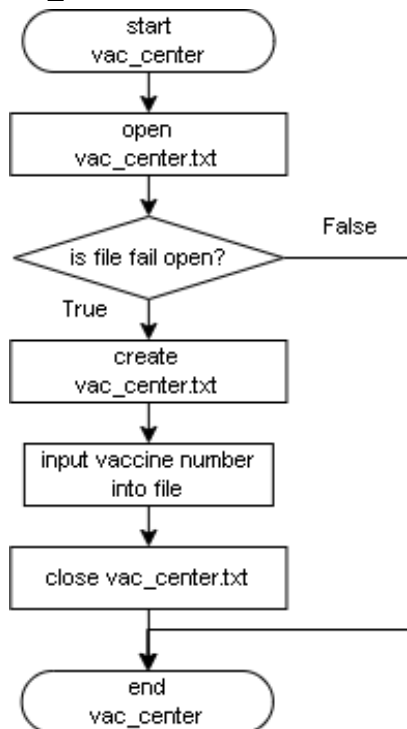
```

Flow Chart

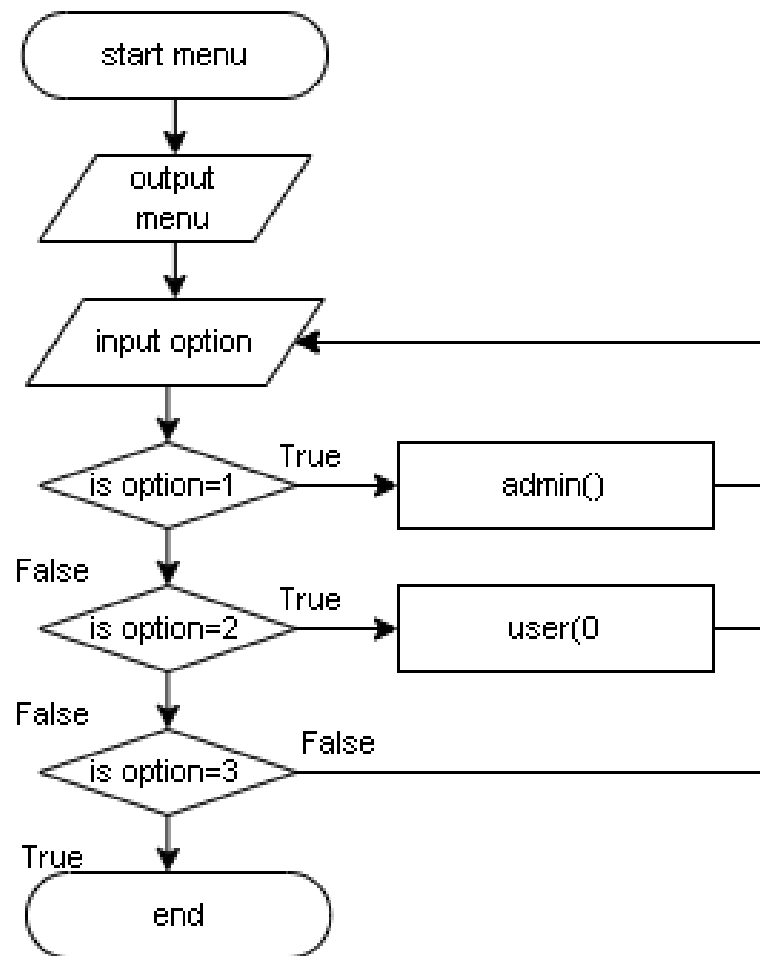
Main flow chart:



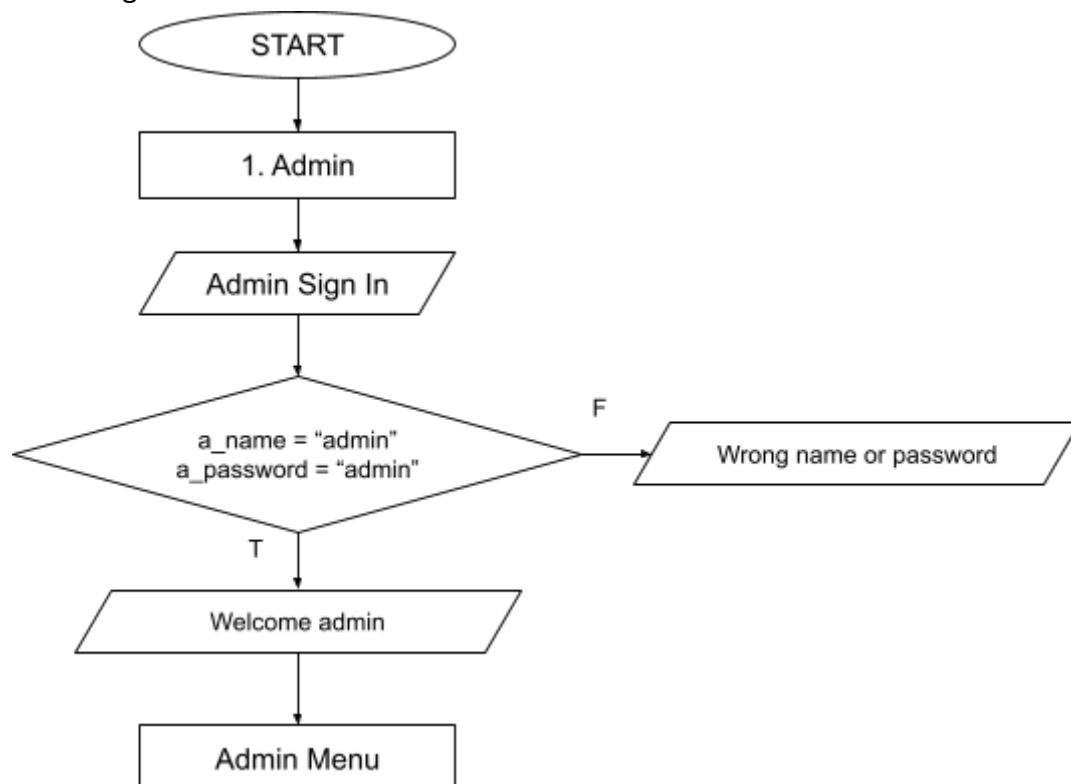
vac_center flow chart:



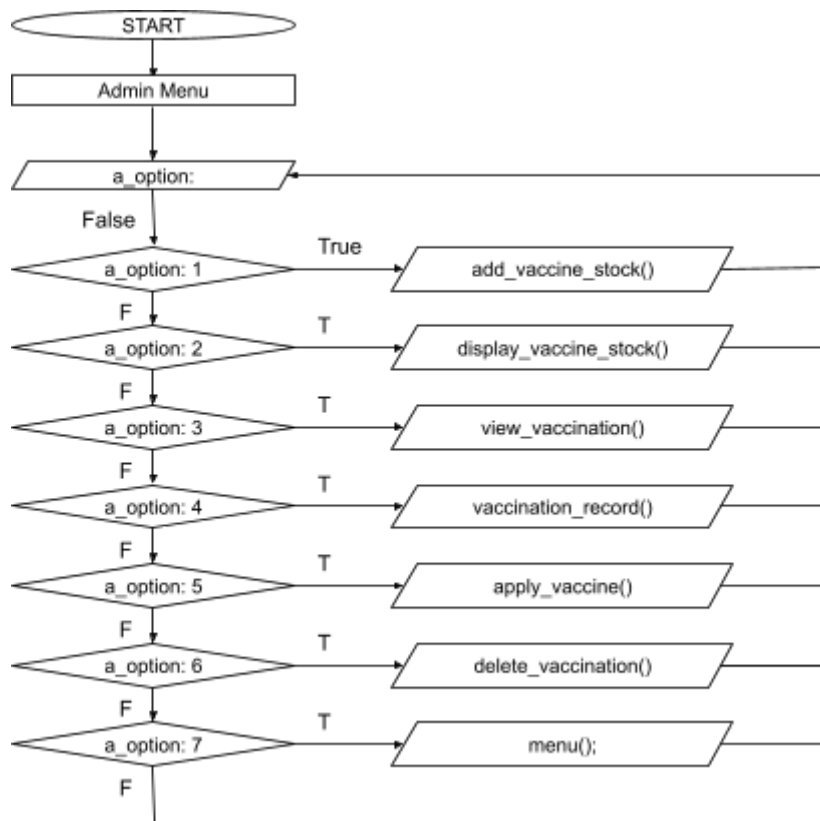
Main menu flow chart:



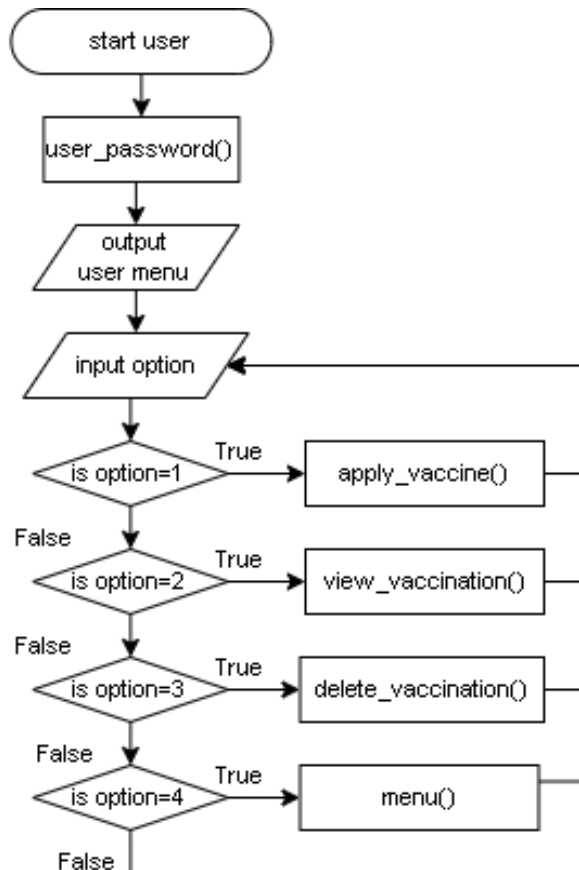
Admin Sign In flow chart:



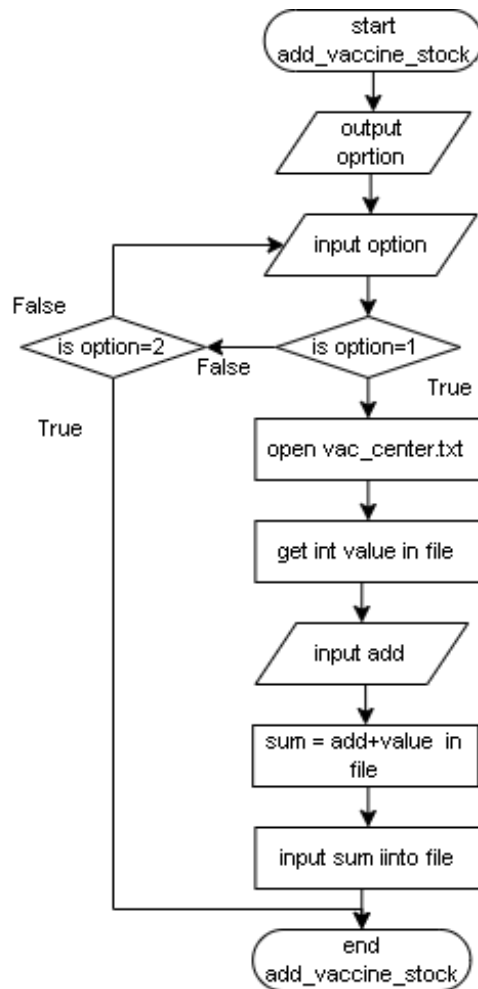
Admin Menu flow chart:



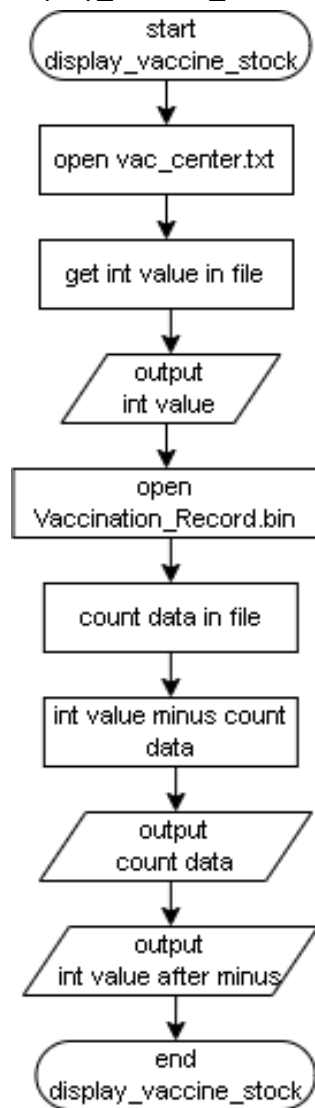
User flow chart:



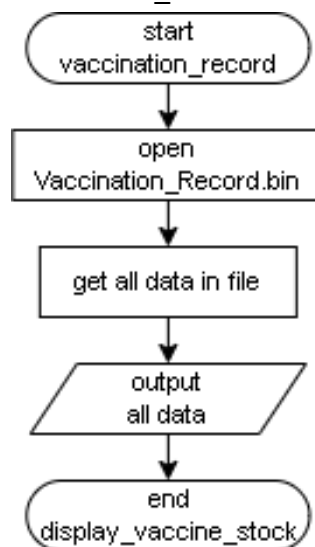
add_vaccine_stock flow chart:



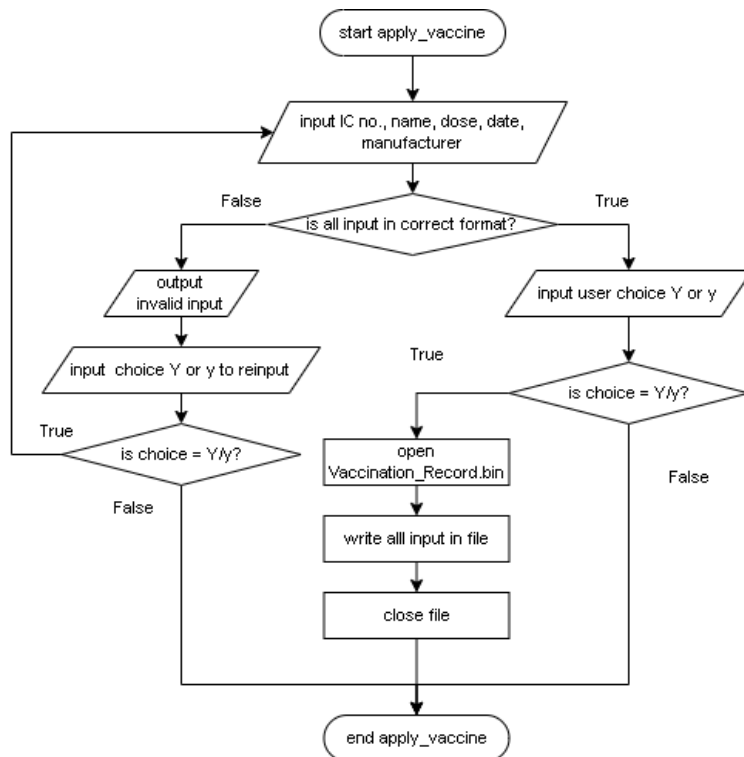
display_vaccine_stock flow chart:



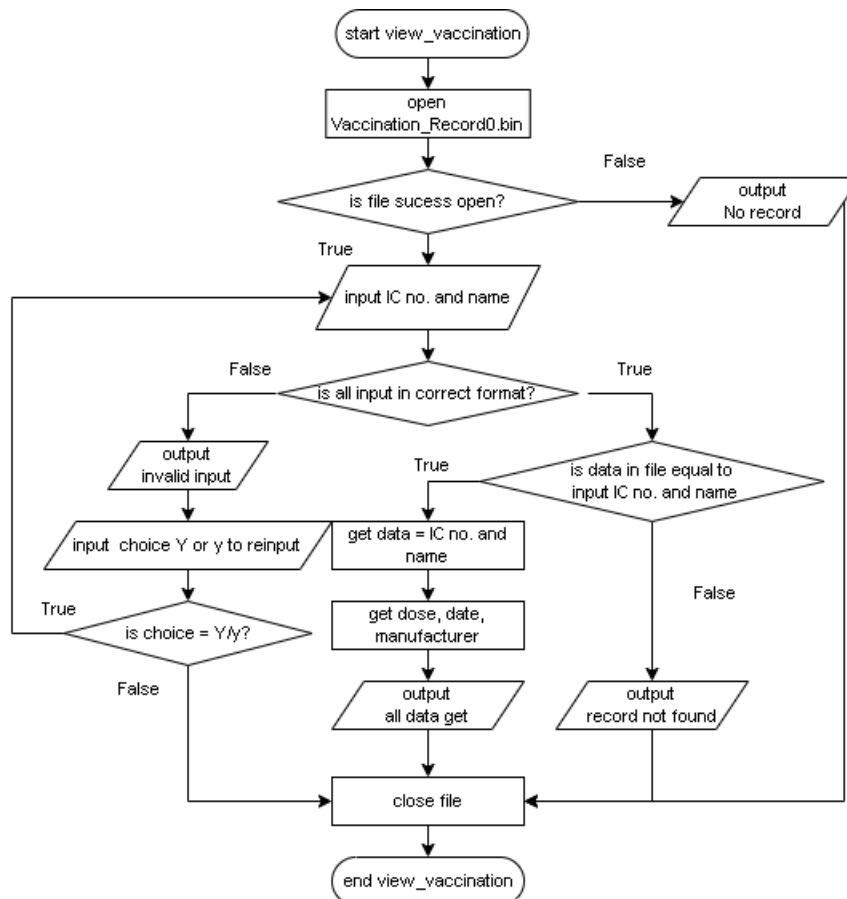
vaccination_record flow chart:



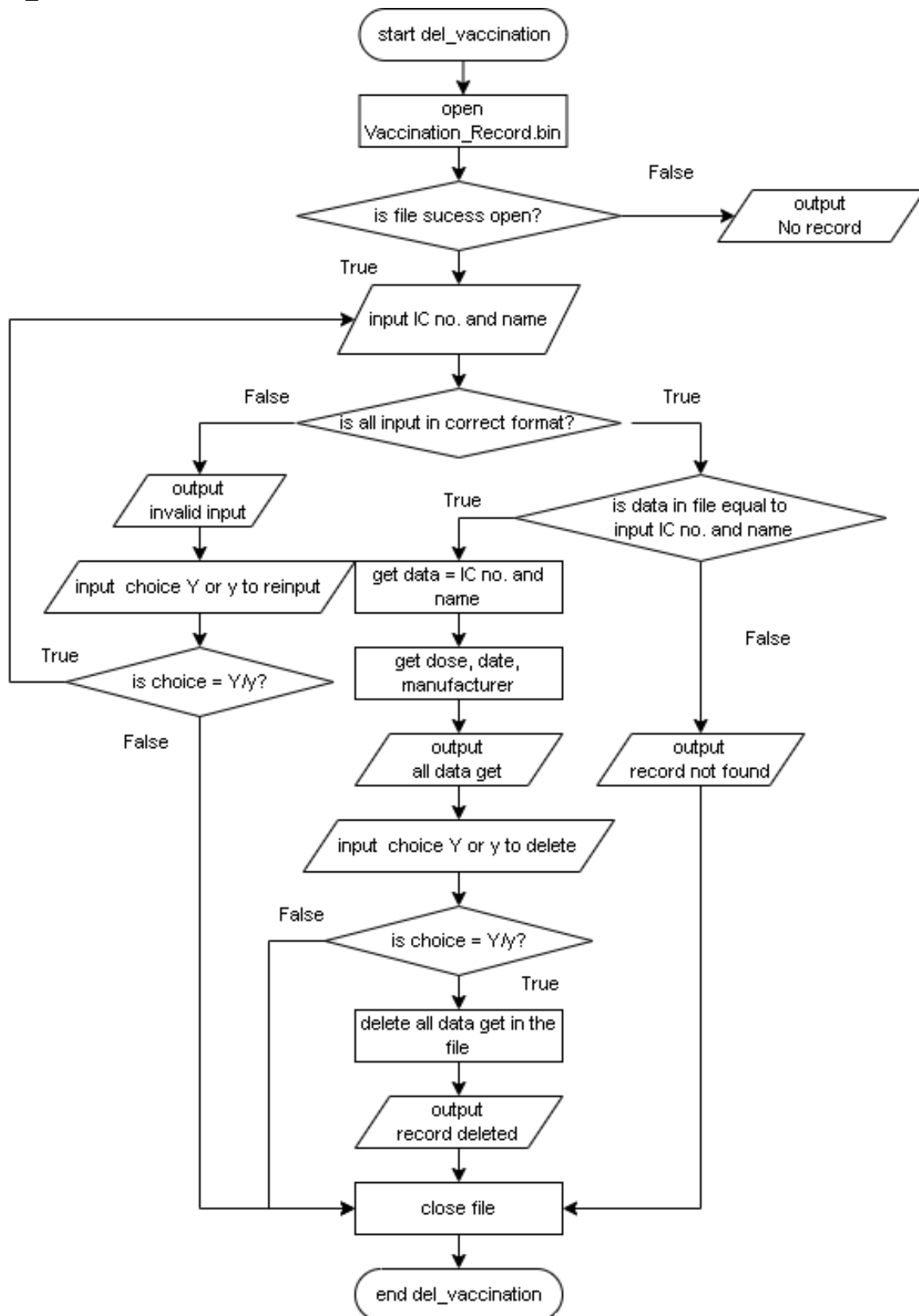
apply_vaccine flow chart:



view_vaccination flow chart:



del_vaccination flow chart:



- **A draft interface design**

Main Menu

COVID VACCINATION SYSTEM

Menu

1. Admin
2. User
3. Exit

Enter Option:

User Sub-Menu

COVID VACCINATION STSTEM

User Sub-Menu

1. User Registration
2. Sign In
3. Change Password
4. Delete Account
5. Back

Enter Option:

User Registration

COVID VACCINATION STSTEM

User Registration

Enter IC No.:

Enter Your Name:

Enter Your Contact number:

Enter Your password:

Sign In User

COVID VACCINATION STSTEM

Sign In User

Enter Your IC No.:

Enter Your Password.:

Change Password (User)

COVID VACCINATION SYSTEM

Change Password

Enter Your IC No. :

Enter Your old Password:

Enter a new password:

Delete Account (User)

COVID VACCINATION SYSTEM

Delete Account

Enter Your IC No. :

Enter Your old Password:

Apply Vaccine (User)

COVID VACCINATION STSTEM

Apply Vaccine

Enter IC No.:

Dose X:

Enter Your Name.:

Enter date (DD/MM/YYYY):

Enter vaccine Manufacture
(Pfizer-BioNTech/Astrazeneca/Sinovac):

Please Double-Check All Information
For Accuracy And Completeness Before Submitting.

Enter Y To Comfirm :

Check Vaccination Record (User)

COVID VACCINATION STSTEM	
Check Vaccination Record	
Enter IC No.:	
Name.:	
Dose X:	
Date :	
Manufacturer:	
Dose X:	
Date:	
Manufacturer:	

Delete Vaccination Record (Admin)

COVID VACCINATION STSTEM	
Delete Vaccination Record	
Enter IC No.:	
Name.:	
Dose X:	
Date:	
Manufacturer:	
Dose X:	
Date:	
Manufacturer:	
Enter Y To Delete Vaccination Record :	

Sign In Admin

COVID VACCINATION STSTEM

Sign In Admin

Name Your Name.:

Enter Your Password:

Show All Vaccination Record (Admin)

COVID VACCINATION STSTEM

Assigned Vaccine Person

IC No.	Name	Dose	Date	Manufacturer
--------	------	------	------	--------------

Show Vaccine Stock (Admin)

COVID VACCINATION STSTEM

Available Vaccine Stock

Vaccine center

Total Vaccine:

Assigned Vaccine Person:

Available Number Of Vaccines are:

Add Vaccine Stock (Admin)

COVID VACCINATION STSTEM

Add Vaccine To Stock

1. Vaccine center

2. Back

Enter Option:

Add Vaccine Stock (Admin) when typed "2"

COVID VACCINATION STSTEM

Add Vaccine To Stock

1. Vaccine center

2. Back

Enter Option:

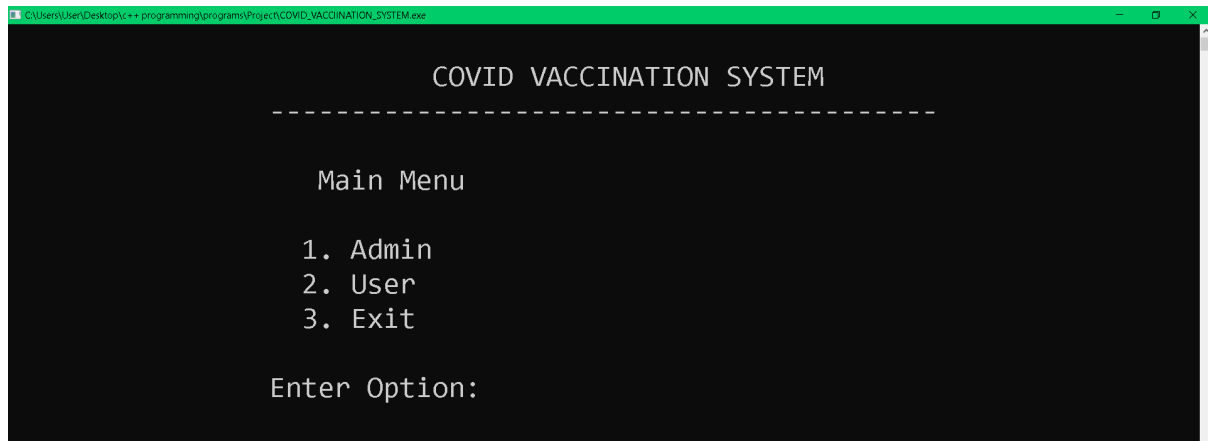
Vaccine In Center:

Enter Number Of Vaccine You Want To Add:

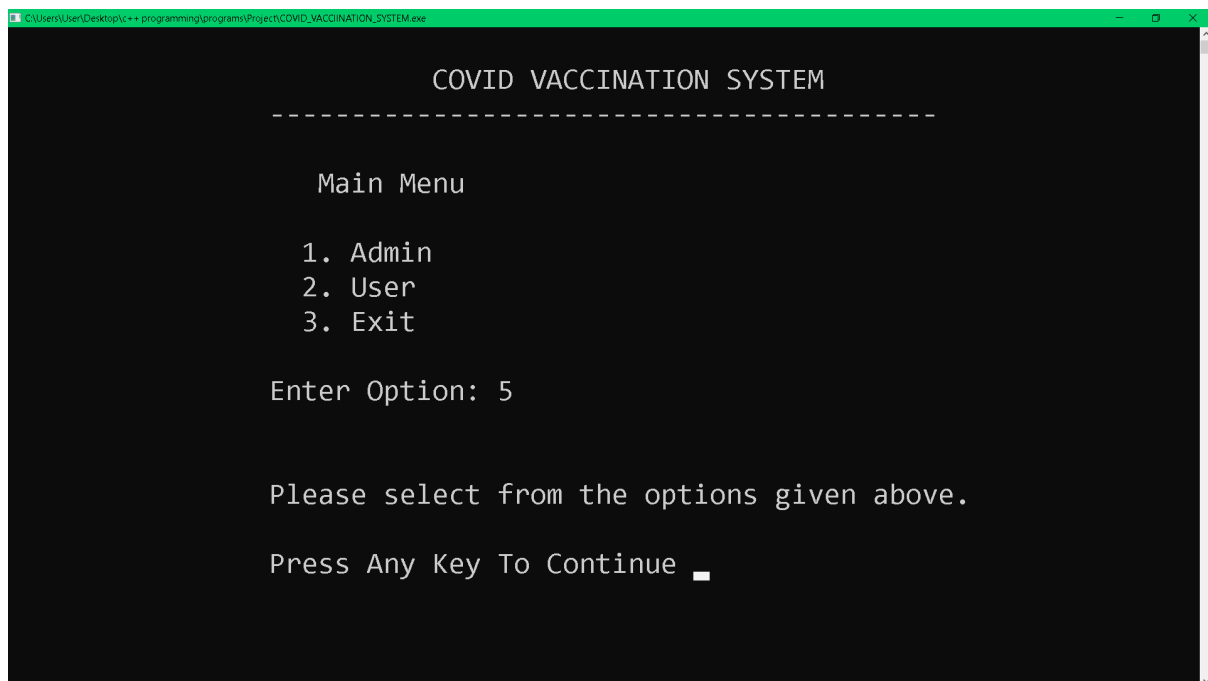
Invalid Input ! Enter A Numeric Value:

Vaccine In Center:

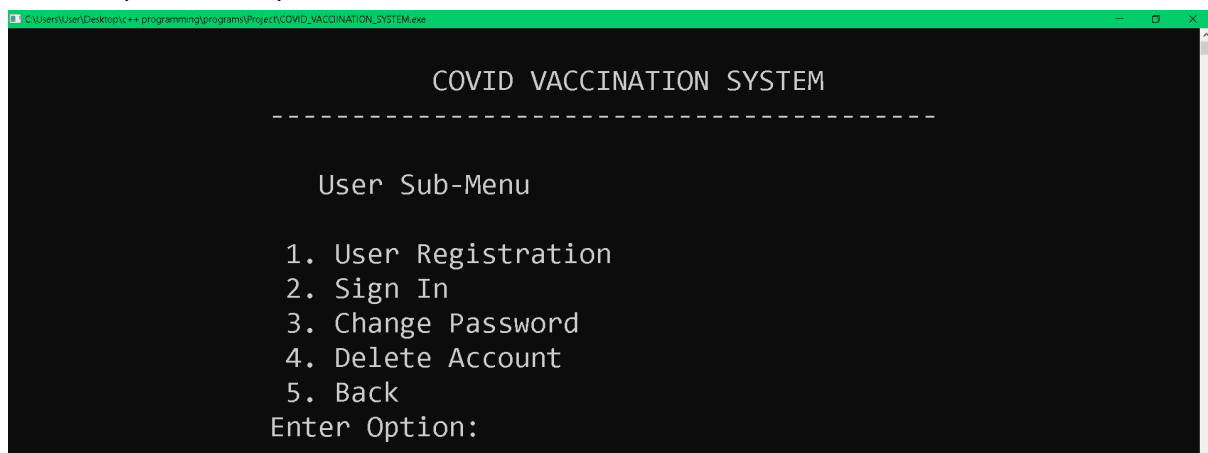
Screenshot of the program



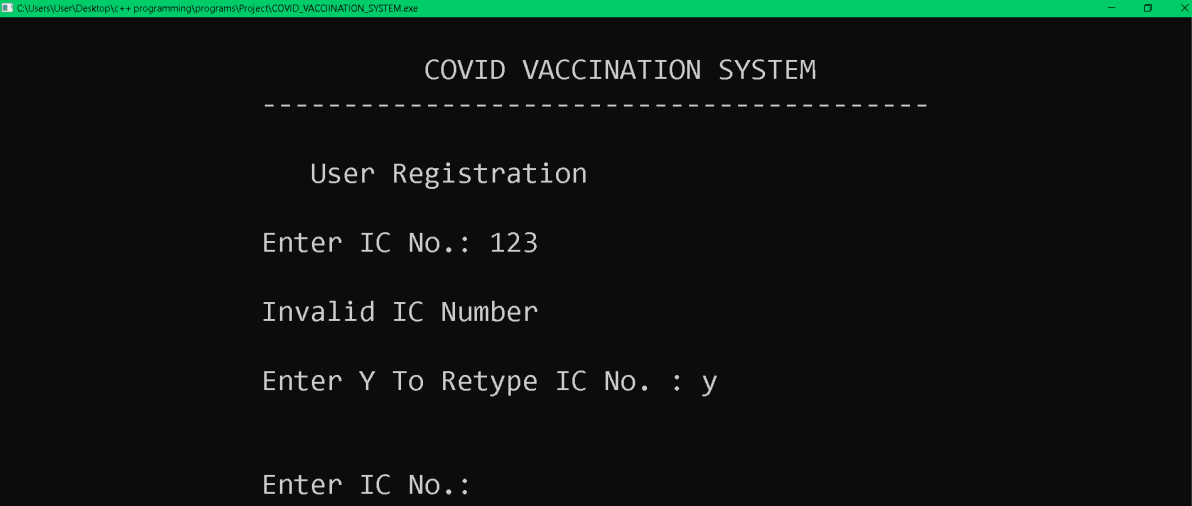
This is the Main Menu, enter option 1, 2 or 3 to the features.



If enter option not equal 1, 2 or 3. User have to reenter.



This is User Sub-Menu, enter option 1 to 5 to the function.



```
COVID VACCINATION SYSTEM
-----

User Registration

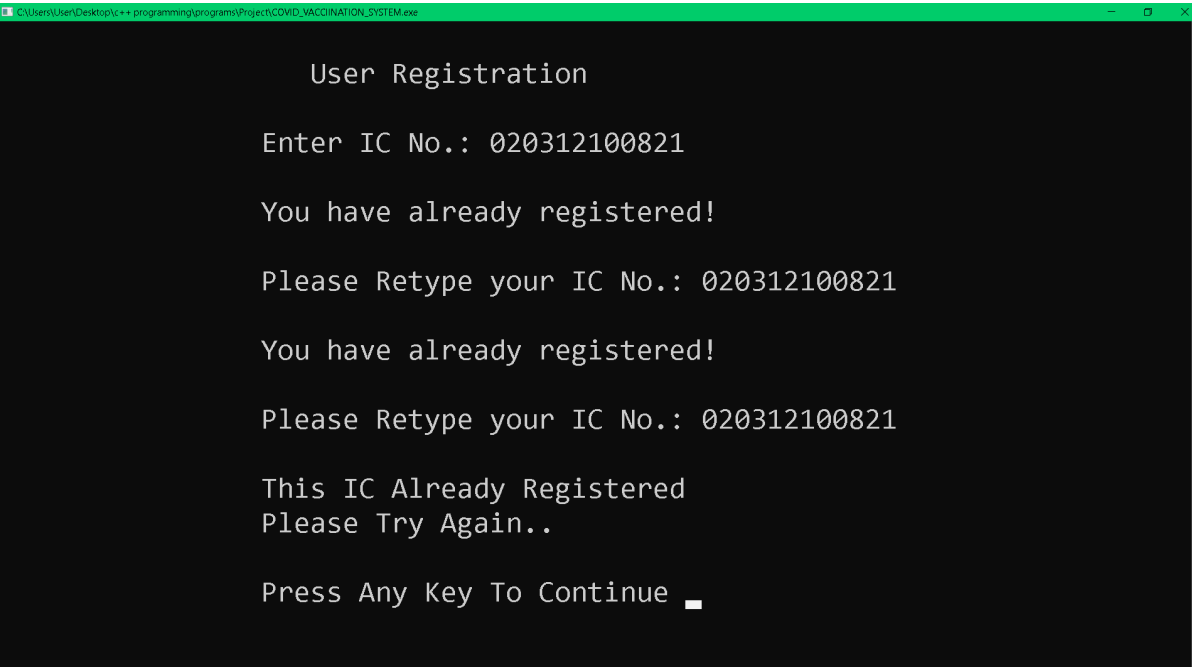
Enter IC No.: 123

Invalid IC Number

Enter Y To Retype IC No. : y

Enter IC No.:
```

If the IC no. lengths not equal to 12, it is not allowed and it will ask whether to retype or exit the User Registration. This parameter applies to all Enter IC no.



```
User Registration

Enter IC No.: 020312100821

You have already registered!

Please Retype your IC No.: 020312100821

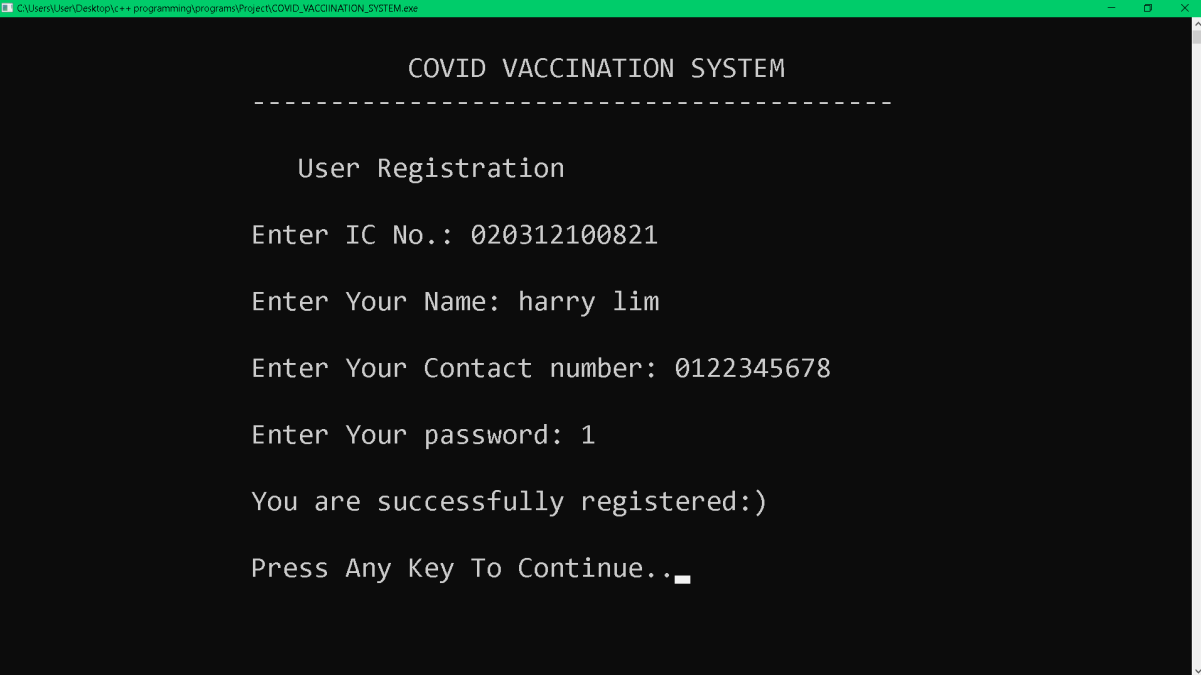
You have already registered!

Please Retype your IC No.: 020312100821

This IC Already Registered
Please Try Again..

Press Any Key To Continue
```

If the IC no. has been registered before. It is not allowed and it will ask to retype the IC number.



```
C:\Users\User\Desktop\c++ programming\programs\Project\COVID_VACCINATION_SYSTEM.exe

COVID VACCINATION SYSTEM
-----

User Registration

Enter IC No.: 020312100821

Enter Your Name: harry lim

Enter Your Contact number: 0122345678

Enter Your password: 1

You are successfully registered:)

Press Any Key To Continue..
```

If user enters all information to the requirement, User is successfully registered.



```
C:\Users\User\Desktop\c++ programming\programs\Project\COVID_VACCINATION_SYSTEM.exe

COVID VACCINATION SYSTEM
-----

Sign In User

Enter Your IC No.: 020312100821

Enter Your Password: 2

Wrong Password

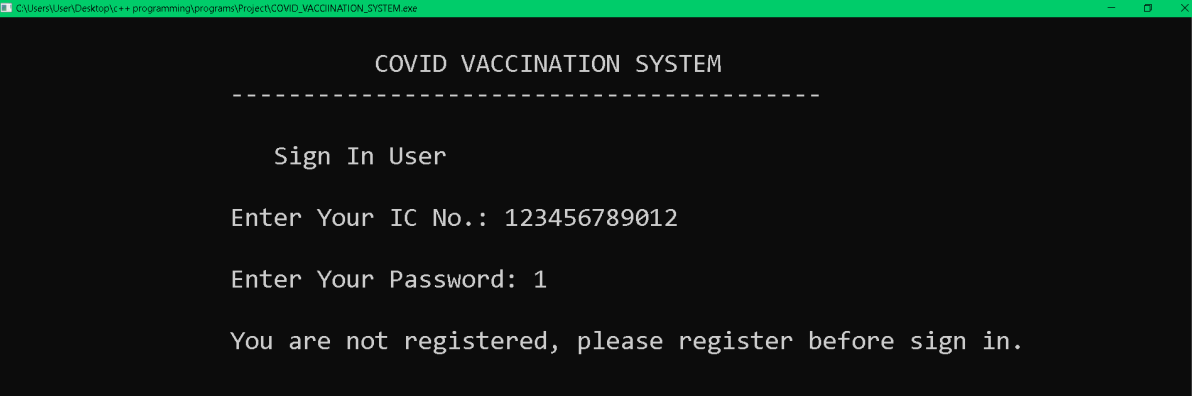
Enter Y To Retype Password : y

Enter Your Password: 1

You are successfully sign in

Press Any Key To Continue..
```

In the Sign In User, the user is required to enter their IC no. and the corresponding password to log in.



```
C:\Users\User\Desktop\c++ programming\programs\Project\COVID_VACCINATION_SYSTEM.exe

COVID VACCINATION SYSTEM
-----

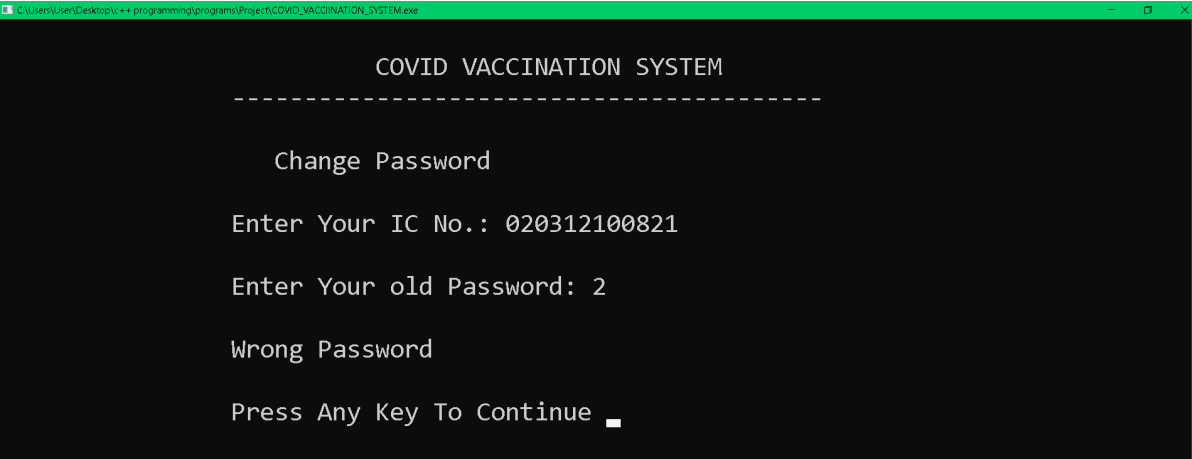
Sign In User

Enter Your IC No.: 123456789012

Enter Your Password: 1

You are not registered, please register before sign in.
```

If the enter IC no. is not registered before, it will ask the user to register first.



```
C:\Users\User\Desktop\c++ programming\programs\Project\COVID_VACCINATION_SYSTEM.exe

COVID VACCINATION SYSTEM
-----

Change Password

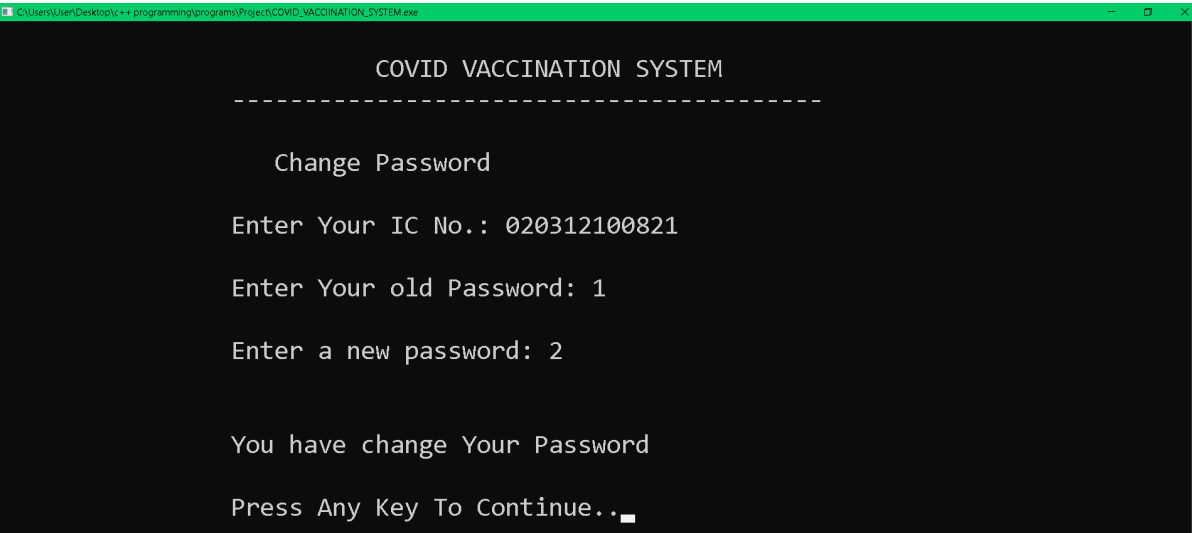
Enter Your IC No.: 020312100821

Enter Your old Password: 2

Wrong Password

Press Any Key To Continue █
```

If a user wants to change their password, the user is required to enter their IC no. and corresponding password before changing their password. Same goes to Delete Account.



```
C:\Users\User\Desktop\c++ programming\programs\Project\COVID_VACCINATION_SYSTEM.exe

COVID VACCINATION SYSTEM
-----

Change Password

Enter Your IC No.: 020312100821

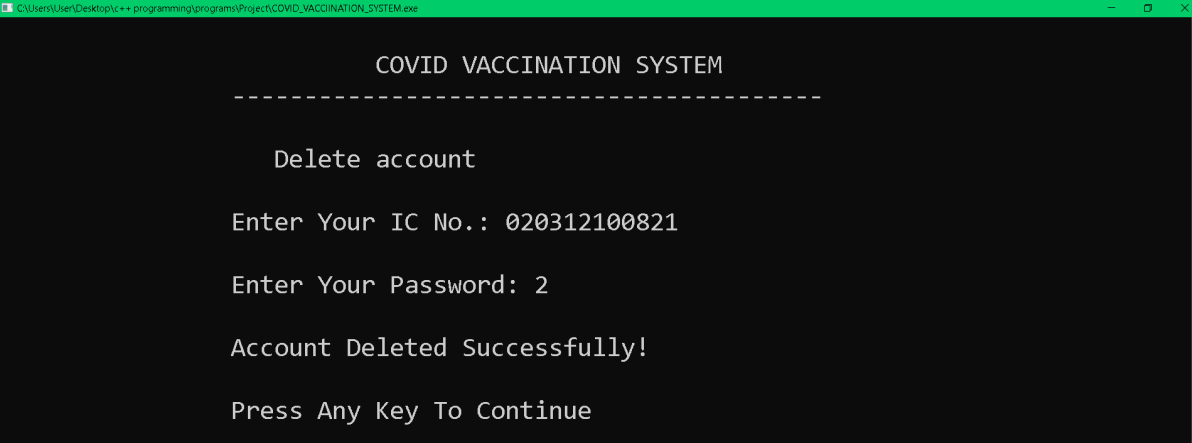
Enter Your old Password: 1

Enter a new password: 2

You have change Your Password

Press Any Key To Continue... █
```

This is when user success changes their password.



```
C:\Users\User\Desktop\c++ programming\programs\Project\COVID_VACCINATION_SYSTEM.exe

COVID VACCINATION SYSTEM
-----

Delete account

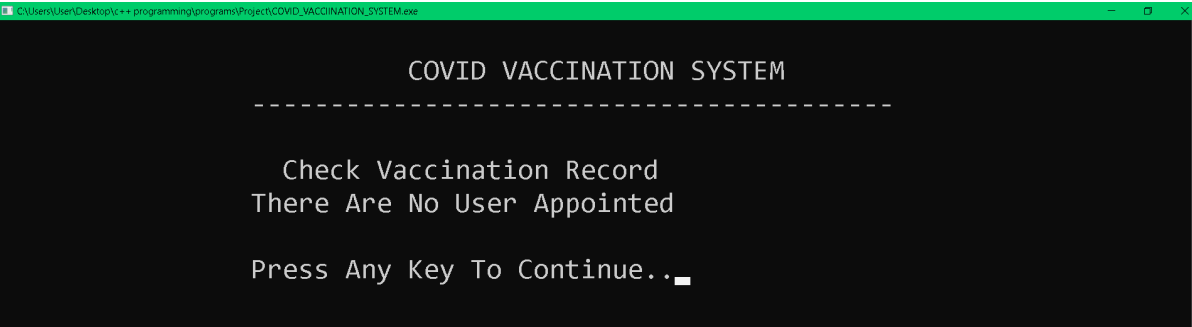
Enter Your IC No.: 020312100821

Enter Your Password: 2

Account Deleted Successfully!

Press Any Key To Continue
```

This is when user success deletes thor account.



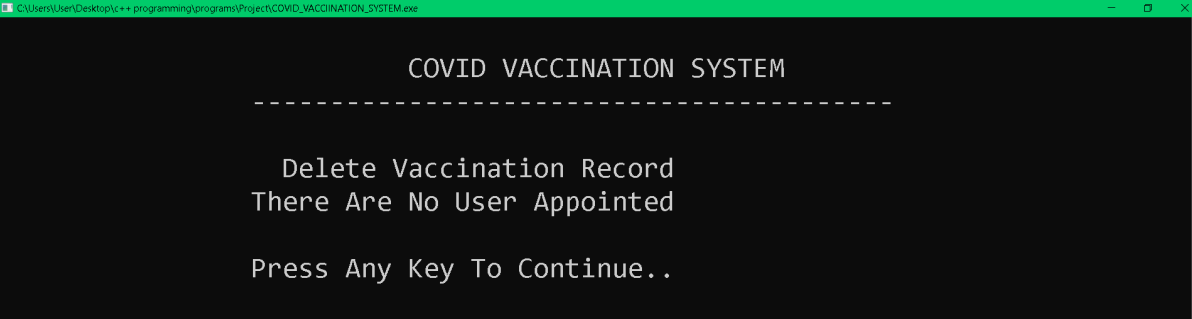
```
C:\Users\User\Desktop\c++ programming\programs\Project\COVID_VACCINATION_SYSTEM.exe

COVID VACCINATION SYSTEM
-----

Check Vaccination Record
There Are No User Appointed

Press Any Key To Continue..
```

In the Check Vaccination Record, if there is no user applied for the vaccine before, it will show "There Are No User Appointer", same with Delete Vaccination Record.

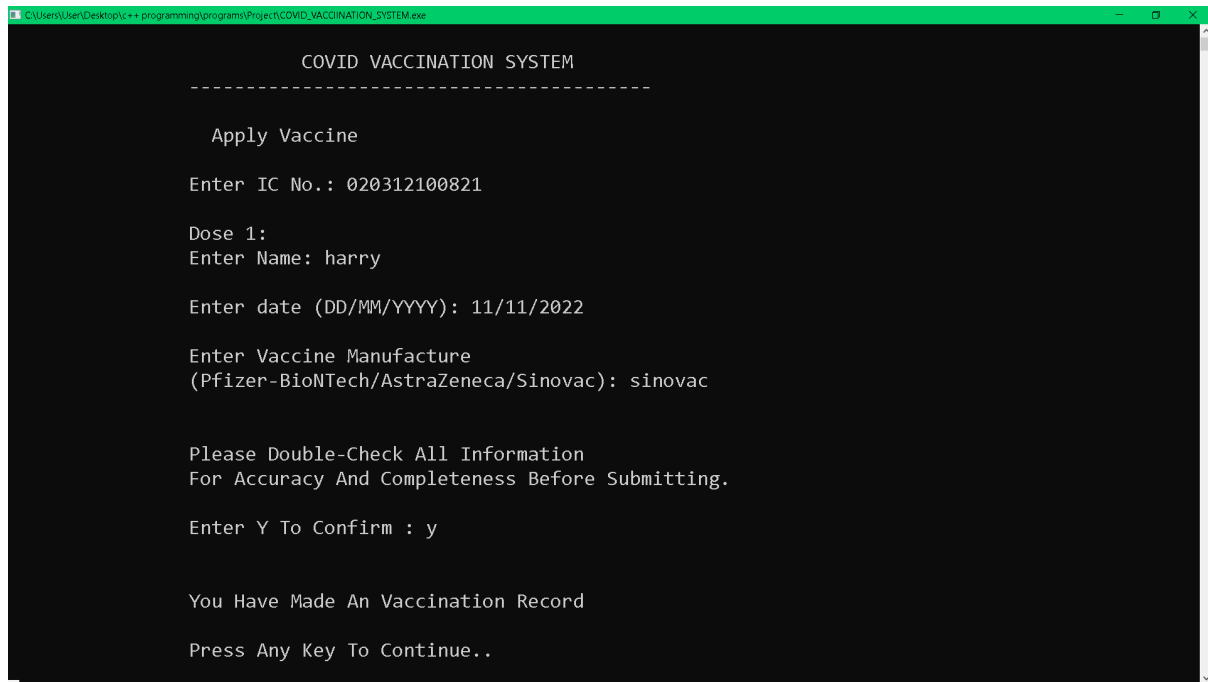


```
C:\Users\User\Desktop\c++ programming\programs\Project\COVID_VACCINATION_SYSTEM.exe

COVID VACCINATION SYSTEM
-----

Delete Vaccination Record
There Are No User Appointed

Press Any Key To Continue..
```



```
COVID VACCINATION SYSTEM
-----

Apply Vaccine

Enter IC No.: 020312100821

Dose 1:
Enter Name: harry

Enter date (DD/MM/YYYY): 11/11/2022

Enter Vaccine Manufacture
(Pfizer-BioNTech/AstraZeneca/Sinovac): sinovac

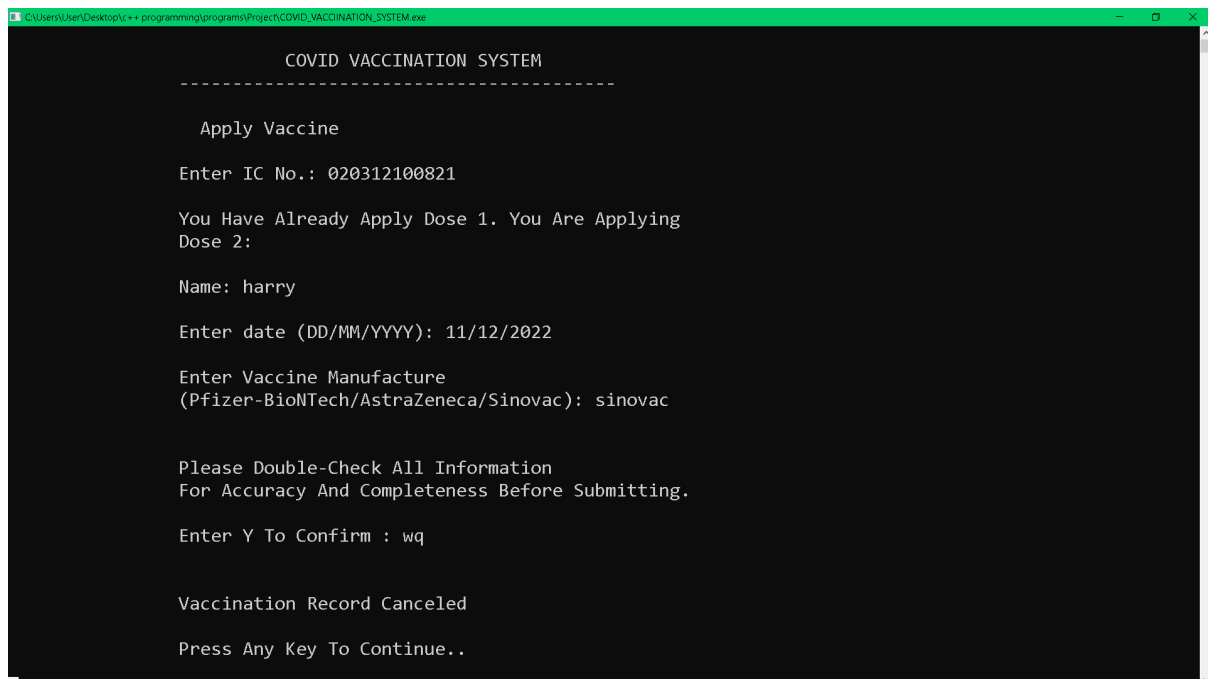
Please Double-Check All Information
For Accuracy And Completeness Before Submitting.

Enter Y To Confirm : y

You Have Made An Vaccination Record

Press Any Key To Continue..
```

In the Apply Vaccine, user is require to enter all the information accordingly and it will ask for a confirmation before record added.



```
COVID VACCINATION SYSTEM
-----

Apply Vaccine

Enter IC No.: 020312100821

You Have Already Apply Dose 1. You Are Applying
Dose 2:

Name: harry

Enter date (DD/MM/YYYY): 11/12/2022

Enter Vaccine Manufacture
(Pfizer-BioNTech/AstraZeneca/Sinovac): sinovac

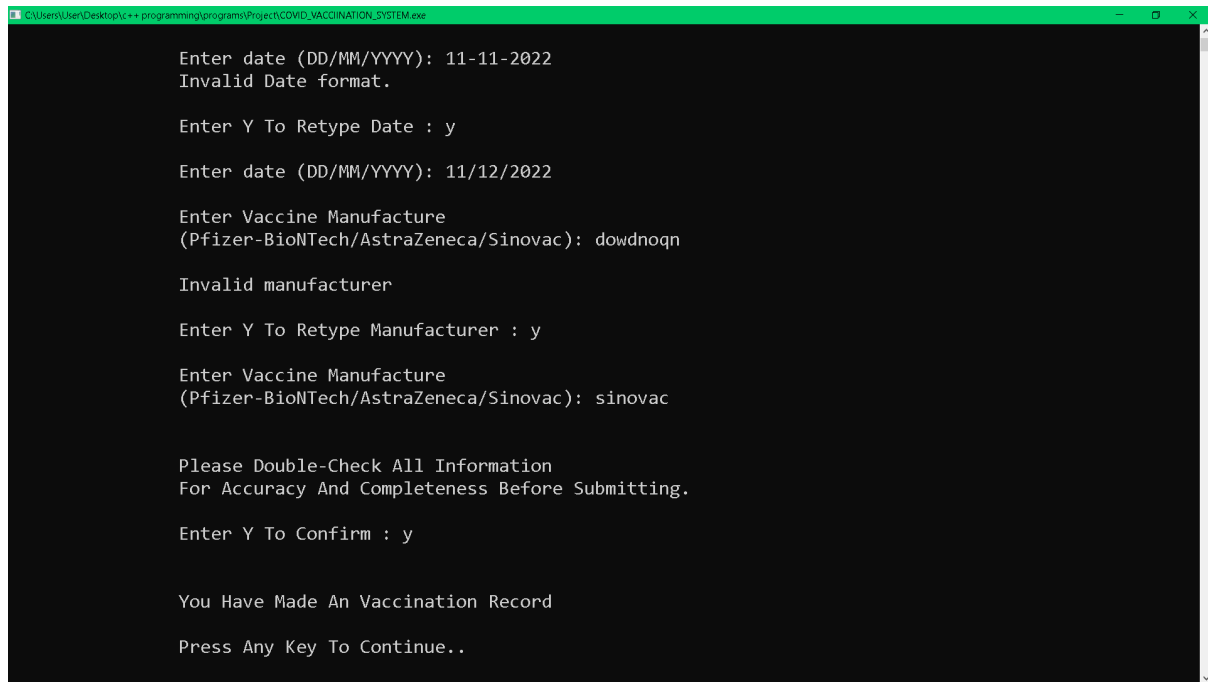
Please Double-Check All Information
For Accuracy And Completeness Before Submitting.

Enter Y To Confirm : wq

Vaccination Record Canceled

Press Any Key To Continue..
```

If the uIC no. has already made the record before, It will notice the user is applying the next dose and show the corresponding name.



```
Enter date (DD/MM/YYYY): 11-11-2022
Invalid Date format.

Enter Y To Retype Date : y

Enter date (DD/MM/YYYY): 11/12/2022

Enter Vaccine Manufacture
(Pfizer-BioNTech/AstraZeneca/Sinovac): dowdnoqn

Invalid manufacturer

Enter Y To Retype Manufacturer : y

Enter Vaccine Manufacture
(Pfizer-BioNTech/AstraZeneca/Sinovac): sinovac

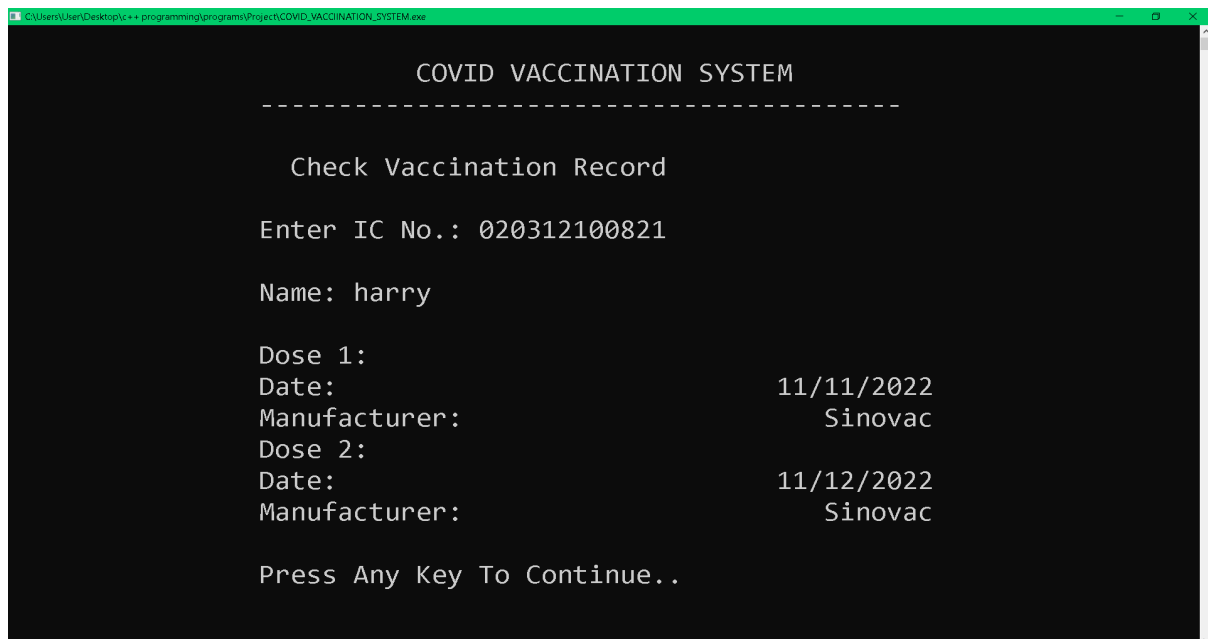
Please Double-Check All Information
For Accuracy And Completeness Before Submitting.

Enter Y To Confirm : y

You Have Made An Vaccination Record

Press Any Key To Continue..
```

If the user enters the wrong date format and manufacture, it will ask the user whether to retype or exit Apply Vaccine.



```
COVID VACCINATION SYSTEM
-----

Check Vaccination Record

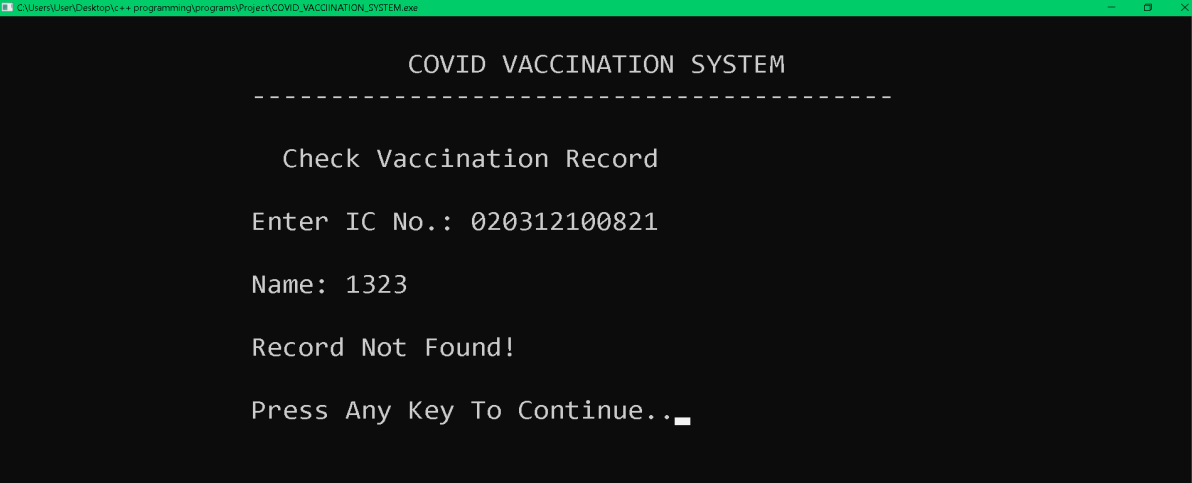
Enter IC No.: 020312100821

Name: harry

Dose 1:
Date: 11/11/2022
Manufacturer: Sinovac
Dose 2:
Date: 11/12/2022
Manufacturer: Sinovac

Press Any Key To Continue..
```

In Check Vaccination Record, the user is required to enter the IC no. and corresponding name apply for vaccine in order to view the record. Same goes for the Delete Vaccination Record.



```
C:\Users\User\Desktop\c++ programming\programs\Project\COVID_VACCINATION_SYSTEM.exe

COVID VACCINATION SYSTEM
-----

Check Vaccination Record

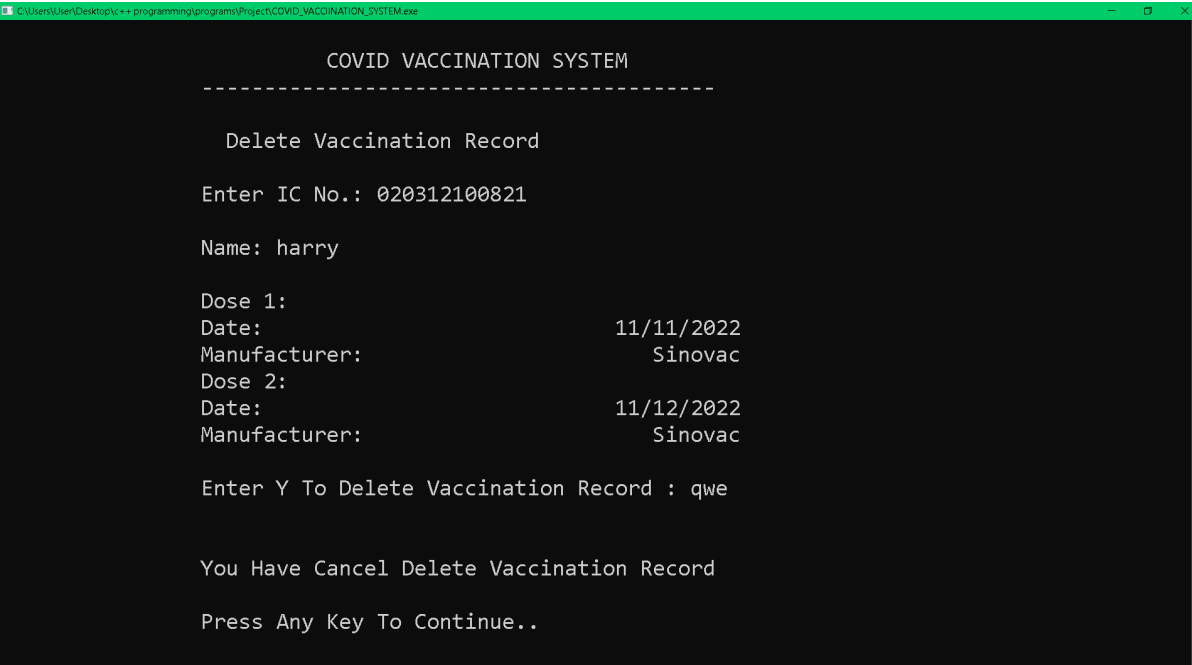
Enter IC No.: 020312100821

Name: 1323

Record Not Found!

Press Any Key To Continue..
```

If the user enters a registered IC no. but the name is not correct, the record will not be shown. Same goes for the Delete Vaccination Record.



```
C:\Users\User\Desktop\c++ programming\programs\Project\COVID_VACCINATION_SYSTEM.exe

COVID VACCINATION SYSTEM
-----

Delete Vaccination Record

Enter IC No.: 020312100821

Name: harry

Dose 1:
Date: 11/11/2022
Manufacturer: Sinovac
Dose 2:
Date: 11/12/2022
Manufacturer: Sinovac

Enter Y To Delete Vaccination Record : qwe

You Have Cancel Delete Vaccination Record

Press Any Key To Continue..
```

In Delete Vaccination Record after the record is shown, it will ask for a confirmation whether to delete the record.

```
C:\Users\User\Desktop\c++ programming\programs\Project\COVID_VACCINATION_SYSTEM.exe

COVID VACCINATION SYSTEM
-----

Delete Vaccination Record

Enter IC No.: 020312100821

Name: harry

Dose 1:
Date: 11/11/2022
Manufacturer: Sinovac
Dose 2:
Date: 11/12/2022
Manufacturer: Sinovac

Enter Y To Delete Vaccination Record : y

You Have Delete The Vaccination Record

Press Any Key To Continue..
```

```
C:\Users\User\Desktop\c++ programming\programs\Project\COVID_VACCINATION_SYSTEM.exe

COVID VACCINATION SYSTEM
-----

Check Vaccination Record

Enter IC No.: 020312100821

Name: harry

Record Not Found!

Press Any Key To Continue..
```

If the user deletes the record. The user is unable to check their record again because it is already deleted.

```
C:\Users\User\Desktop\c++ programming\programs\Project\COVID_VACCINATION_SYSTEM.exe

COVID VACCINATION SYSTEM
-----

Sign In Admin

Enter Your Name: weifnwi2

Enter Your Password: 12eini

Wrong Name Or Password.
```

In the Sign In admin, admin is required to “admin” for the name and admin for the password. Else, it will not sign in to Admin Menu.

```
COVID VACCINATION SYSTEM
-----

Sign In Admin

Enter Your Name: admin

Enter Your Password: 1238h8

Wrong Name Or Password.
```

```
COVID VACCINATION SYSTEM
-----

Sign In Admin

Enter Your Name: admin

Enter Your Password: admin

Welcome Admin.
```

```
COVID VACCINATION SYSTEM
-----

Assigned Vaccine Person

IC No.      Name      Dose Date      Manufacturer
020312100821  harry    1    11/11/2022  Sinovac
020312100821  harry    2    11/12/2022  Sinovac
020823100523  Jonny    1    12/12/2022  Pfizer-BioNTech

Press Any Key To Continue..
```

In Assigned Vaccine Person, all IC no. apply for vaccine will be shown.

```
C:\Users\User\Desktop\c++ programming\programs\Project\COVID_VACCINATION_SYSTEM.exe

      COVID VACCINATION SYSTEM
      -----

      Available Vaccine Stock

      Vaccine Center
      Total Vaccine: 50000

      Assigned Vaccine Person: 3

      Available Number Of Vaccines are: 49997

      Press Any Key To Continue..
```

In Available Stock, the vaccine in center is shown and it will minus the assigned vaccine person for the available vaccine number.

```
C:\Users\User\Desktop\c++ programming\programs\Project\COVID_VACCINATION_SYSTEM.exe

      COVID VACCINATION SYSTEM
      -----

      Add Vaccine To Center

      1. Vaccine Center
      2. Back

      Enter Option: █
```

In Add Vaccine to Center, admin enter 1 to add vaccine, 2 to back to admin menu. If user enter 1, vaccine in center is shown and user is required to enter number only.

```
C:\Users\User\Desktop\c++ programming\programs\Project\COVID_VACCINATION_SYSTEM.exe

      Add Vaccine To Center

      1. Vaccine Center
      2. Back

      Enter Option: 1

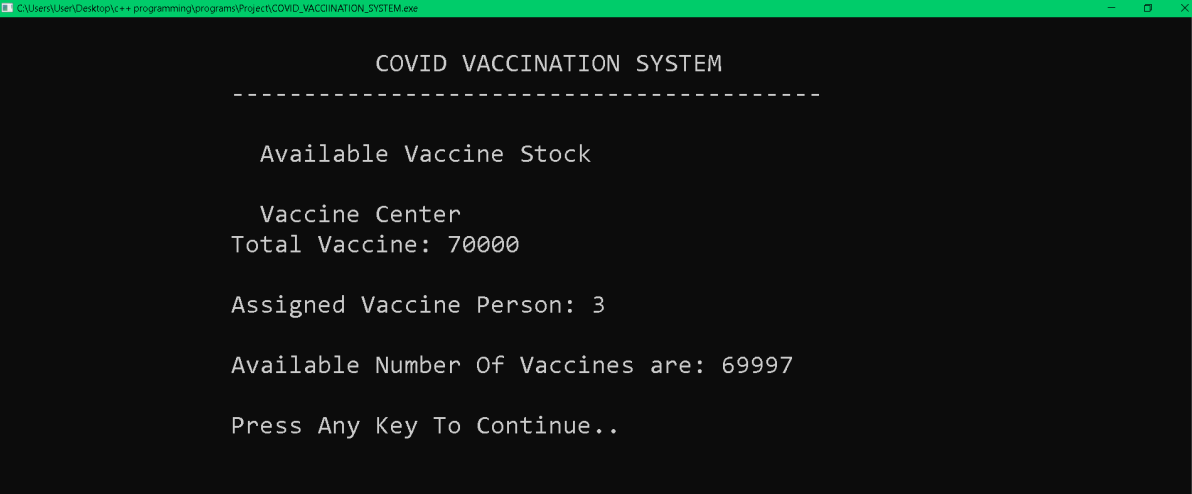
      Vaccine In Center: 50000
      Enter Number Of Vaccines You Want To Add: wd9dw

      Invalid Input! Enter A Numeric Value: 20000

      Vaccine In Center: 70000

      Vaccine Successfully Added

      Press Any Key To Continue..
```



```
C:\Users\User\Desktop\c++ programming\programs\Project\COVID_VACCINATION_SYSTEM.exe

COVID VACCINATION SYSTEM
-----

Available Vaccine Stock

Vaccine Center
Total Vaccine: 70000

Assigned Vaccine Person: 3

Available Number Of Vaccines are: 69997

Press Any Key To Continue..
```

This is to show the vaccine is successfully added to the system.

Reference

1. Ulla Kirch-Prinz and Peter Prinz, (2002) A Complete Guide to Programming in C++. Jones and Bartlett Publishers. Retrieved from <http://www.lmpt.univ-tours.fr/~volkov/C++.pdf>
2. StudiesToday. CBSE Class 12 Computer Science - Data File Handling In C++. StudiesToday.com. Retrieved from <https://www.studiestoday.com/concept-computer-science-cbse-class-12-computer-science-data-file-handling-c-220068.html>

Appendix

```

#include <iostream>
#include <cstring> // strcpy compare string
#include <windows.h> // functions in the Windows API
#include <fstream> // create files, write information to files, and read information from files.
#include <conio.h> // getch
#include <iomanip> // setfill, setw
#include <string>
#include <cstdlib>
#include <limits>

using namespace std;

class covid_vaccination
{
protected:
    string name;
    string password;
    string contact;
    string user_name, user_IC;
    string IC;

    // vac center
    string VacCenter = "Vaccine Center";

    // vaccination data
    struct vac_data
    {
        char ulC[13];
        char uname[100];
        char dose;
        char manufacturer[16];
        char udate[11];
    };
    int counter;

public:
    void vac_center();

    void title();
    void retypeOpt(); // retype option

    void menu();

    // admin
    void admin();
    void admin_password();

    // user
    void user();
    void user_password();
    void valid(string str); // check IC exist

```

```
// admin func
void add_vaccine_stock();
void display_vaccine_stock();
void vaccination_record();

// both func
void apply_vaccine();
void view_vaccination();
void delete_vaccination();
};

void covid_vaccination::vac_center()
{
    fstream file;

    file.open("vac_center.txt", ios::in);
    if (file.fail())
    {
        file.close();
        file.open("vac_center.txt", ios::out | ios::app);
        file << 50000;
    }
    file.close();
}

void covid_vaccination::title()
{
    cout << "\n\t\t\t\t\t COVID VACCINATION SYSTEM \t\t";
    cout << "\n\t\t\t-----";
}

void covid_vaccination::retypeOpt()
{
    cout << "\n\n\t\tPlease select from the options given above.";
    cout << "\n\n\t\tPress Any Key To Continue ";
    getch();
    cin.clear();
    fflush(stdin); // clear cin buffer
    cout.flush(); // clear cout buffer
}

void covid_vaccination::menu()
{
    system("cls"); // clear screen
    int option;
    title();
    // main menu
    cout << "\n\n\t\t Main Menu ";
    cout << "\n\n\t\t 1. Admin";
    cout << "\n\t\t 2. User";
    cout << "\n\t\t 3. Exit";
    cout << "\n\n\t\tEnter Option: ";
    cin >> option;
```



```

// call function by chosen num
switch (option)
{
case 1:
    admin();
    break;
case 2:
    user();
    break;
case 3:
    system("cls");
    cout << "\n\n\t\t COVID VACCINATION SYSTEM";
    Sleep(10);
    exit(0);
default:
    retypeOpt();
    menu();
}
}

// admin
void covid_vaccination::admin()
{
    admin_password();
A:
    system("cls");
    int a_option;
    title();
    cout << "\n\n\t\t Admin Menu ";
    cout << "\n\n\t\t 1. Add Vaccine Stock";
    cout << "\n\t\t 2. Show Vaccine Stock";
    cout << "\n\t\t 3. Search Vaccination Record";
    cout << "\n\t\t 4. Show All Vaccination Record";
    cout << "\n\t\t 5. Add Vaccination Record";
    cout << "\n\t\t 6. Delete Vaccination Record";
    cout << "\n\t\t 7. Sign Out";
    cout << "\n\n\t\t Enter Option: ";
    cin >> a_option;
    switch (a_option)
    {
case 1:
        add_vaccine_stock();
        goto A;
        break;
case 2:
        display_vaccine_stock();
        goto A;
        break;
case 3:
        view_vaccination();
        goto A;
        break;
case 4:
        vaccination_record();
    }
}

```

```

        goto A;
        break;
    case 5:
        apply_vaccine();
        goto A;
        break;
    case 6:
        delete_vaccination();
        goto A;
        break;
    case 7:
        menu();
    default:
        retypeOpt();
        goto A;
        break;
    }
}

void covid_vaccination::admin_password()
{
    system("cls");
    char a_name[20];
    char a_password[20];
    title();
    cout << "\n\n\t\t Sign In Admin ";
    cout << "\n\n\t\tEnter Your Name: ";
    cin >> a_name;
    cout << "\n\t\tEnter Your Password: ";
    cin >> a_password;
    if ((strcmp(a_name, "admin") == 0) && (strcmp(a_password, "admin") == 0))
    {
        cout << "\n\n\t\tWelcome Admin.\n\n";
        getch();
        system("cls");
    }
    else
    {
        cout << "\n\n\t\tWrong Name Or Password.\n\n";
        getch();
        system("cls");
        menu();
    }
}

// user
void covid_vaccination::user()
{
    user_password();
B:
    system("cls");
    int u_option;
    title();
    cout << "\n\n\t\t User Menu ";

```

```

cout << "\n\t\t Welcome " << user_name;
cout << "\n\t\t 1. Apply For Vaccine";
cout << "\n\t\t 2. View Vaccination Record";
cout << "\n\t\t 3. Delete Vaccination Record";
cout << "\n\t\t 4. Sign Out";
cout << "\n\n\t\t Enter Option: ";
cin >> u_option;
switch (u_option)
{
case 1:
    apply_vaccine();
    goto B;
case 2:
    view_vaccination();
    goto B;
case 3:
    delete_vaccination();
    goto B;
case 4:
    menu();
default:
    retypeOpt();
    goto B;
}
}

```

```

void covid_vaccination::user_password()
{
    system("cls");
    int u_option;
    string fname, usern, userp;
    ofstream fileo;
    ifstream filei;
    title();
    cout << "\n\n\t\t User Sub-Menu ";
    cout << "\n\n\t\t 1. User Registration";
    cout << "\n\t\t 2. Sign In";
    cout << "\n\t\t 3. Change Password";
    cout << "\n\t\t 4. Delete Account";
    cout << "\n\t\t 5. Back";
    cout << "\n\t\t Enter Option: ";
    cin >> u_option;
    if (u_option == 1)
    {
        system("cls");
        title();
        cout << "\n\n\t\t User Registration ";
A:
        cout << "\n\n\t\t Enter IC No.: ";
        cin >> IC;
        if (IC.length() != 12)
        {
            cout << "\n\t\t Invalid IC Number";
            char choice;

```

```

        cout << "\n\n\t\tEnter Y To Retype IC No. : ";
        cin >> choice;
        if (choice == 'y' || choice == 'Y')
        {
            goto A;
        }
        else
        {
            cin.clear();
            fflush(stdin); // clear input buffer
            user();
        }
    }
    counter = 0;
    valid(IC);
    if (counter >= 3)
    {
        cout << "\n\n\t\tPress Any Key To Continue ";
        getch();
        user_password();
    }
    cout << "\n\n\t\tEnter Your Name: ";
    cin.ignore();
    getline(cin, name);
    cout << "\n\n\t\tEnter Your Contact number: ";
    cin >> contact;
    cout << "\n\n\t\tEnter Your password: ";
    cin >> password;
    fname = IC + ".txt";
    fileo.open(fname.c_str());
    fileo << name << endl
        << password << endl
        << contact << endl;
    cout << "\n\n\t\tYou are successfully registered:>";
    cout << "\n\n\t\tPress Any Key To Continue..";
    Sleep(500);
    getch();
    fileo.close();
    user();
}
else if (u_option == 2)
{
    system("cls");
    string u_IC, u_pass;
    title();
    cout << "\n\n\t\t Sign In User ";
B:
    cout << "\n\n\t\tEnter Your IC No.: ";
    cin >> u_IC;
    if (u_IC.length() != 12)
    {
        cout << "\n\n\t\tInvalid IC Number";
        char choice;
        cout << "\n\n\t\tEnter Y To Retype IC No. : ";
    }
}

```

```

    cin >> choice;
    if (choice == 'y' || choice == 'Y')
    {
        goto B;
    }
    else
    {
        cin.clear();
        fflush(stdin); // clear input buffer
        user();
    }
}
C:
cout << "\n\t\tEnter Your Password: ";
cin >> u_pass;
fname = u_IC + ".txt";
filei.open(fname.c_str());
if (!filei.is_open() && filei.fail())
{
    cout << "\n\t\tYou are not registered, please register before sign in.\n";
    filei.close();
    getch();
    user_password();
}
getline(filei, usern);
getline(filei, userp);
if (u_pass == userp)
{
    user_IC = u_IC;
    user_name = usern;
    cout << "\n\t\tYou are successfully sign in";
    cout << "\n\n\t\tPress Any Key To Continue..";
    getch();
}
else if (u_pass != userp)
{
    cout << "\n\t\tWrong Password";
    char choice2;
    cout << "\n\n\t\tEnter Y To Retype Password : ";
    cin >> choice2;
    if (choice2 == 'y' || choice2 == 'Y')
    {
        goto C;
    }
    else
    {
        cout << "\n\n\t\tPress Any Key To Continue ";
        getch();
        cin.clear();
        fflush(stdin);
        cout.flush();
        user_password();
    }
}
}

```

```

}
else if (u_option == 3)
{
    system("cls");
    string u_IC, u_pass;
    title();
    cout << "\n\n\t\t Change Password ";
D:
    cout << "\n\n\t\tEnter Your IC No.: ";
    cin >> u_IC;
    if (u_IC.length() != 12)
    {
        cout << "\n\t\tInvalid IC Number";
        char choice;
        cout << "\n\n\t\tEnter Y To Retype IC No. : ";
        cin >> choice;
        if (choice == 'y' || choice == 'Y')
        {
            goto D;
        }
        else
        {
            cin.clear();
            fflush(stdin); // clear input buffer
            user();
        }
    }
    fname = u_IC + ".txt";
    filei.open(fname.c_str());
    if (!filei.is_open() && filei.fail())
    {
        cout << "\n\t\tYou are not registered, please register before changing password.\n";
        filei.close();
        getch();
        user_password();
    }
    getline(filei, usern);
    getline(filei, userp);
    filei.close();
    cout << "\n\t\tEnter Your old Password: ";
    cin >> u_pass;
    if (u_pass == userp)
    {
        string newPassword;
        cout << "\n\t\tEnter a new password: ";
        cin >> newPassword;
        ofstream fileo;
        fileo.open(fname.c_str());
        fileo << usern << endl
            << newPassword;
        fileo.close();
        cout << "\n\n\t\tYou have change Your Password";
        cout << "\n\n\t\tPress Any Key To Continue..";
        getch();
    }
}

```

```

        system("cls");
        cin.clear();
        fflush(stdin);
        cout.flush();
        user_password();
    }
    else if (u_pass != userp)
    {
        cout << "\n\t\tWrong Password";
        cout << "\n\n\t\tPress Any Key To Continue ";
        getch();
        cin.clear();
        fflush(stdin);
        cout.flush();
        user_password();
    }
}
else if (u_option == 4)
{
    system("cls");
    string u_IC, u_pass;
    title();
    cout << "\n\n\t\t Delete account ";
E:
    cout << "\n\n\t\tEnter Your IC No.: ";
    cin >> u_IC;
    if (u_IC.length() != 12)
    {
        cout << "\n\t\tInvalid IC Number";
        char choice;
        cout << "\n\n\t\tEnter Y To Retype IC No. : ";
        cin >> choice;
        if (choice == 'y' || choice == 'Y')
        {
            goto E;
        }
        else
        {
            cin.clear();
            fflush(stdin); // clear input buffer
            user();
        }
    }
}
fname = u_IC + ".txt";
filei.open(fname.c_str());
if (!filei.is_open() && filei.fail())
{
    cout << "\n\t\tAccount not exist.\n";
    filei.close();
    getch();
    user_password();
}
getline(filei, usern);
getline(filei, userp);

```

```

    filei.close();
    cout << "\n\t\tEnter Your Password: ";
    cin >> u_pass;
    if (u_pass == userp)
    {
        int result = remove(fname.c_str());
        if (result == 0)
        {
            cout << "\n\t\tAccount Deleted Successfully!";
            cout << "\n\n\t\tPress Any Key To Continue ";
            getch();
            cin.clear();
            fflush(stdin);
            cout.flush();
            menu();
        }
        else
        {
            cout << "\n\t\tError Occurred!";
            cout << "\n\n\t\tPress Any Key To Continue ";
            getch();
            cin.clear();
            fflush(stdin);
            cout.flush();
            user_password();
        }
    }
}
else if (u_option == 5)
{
    menu();
}
else
{
    retypeOpt();
    user_password();
}
}

void covid_vaccination::valid(string str) // Check Username exist
{
    string dir, user;
    ifstream file;
    dir = str + ".txt";
    file.open(dir.c_str());           // converts a string to an array of characters with a null character at the
end.
    if (!file.is_open() && file.fail()) // f
    {
        return;
    }
    else
    {
        counter++;
        if (counter >= 3)

```



```

    {
        cout << "\n\t\tThis IC Already Registered\n\t\tPlease Try Again..";
        file.close();
        return;
    }
    cout << "\n\t\tYou have already registered!";
B:
    cout << "\n\n\t\tPlease Retype your IC No.: ";
    cin >> IC;
    if (IC.length() != 12)
    {
        cout << "\n\t\tInvalid IC Number";
        char choice;
        cout << "\n\n\t\tEnter Y To Retype IC No. : ";
        cin >> choice;
        if (choice == 'y' || choice == 'Y')
        {
            goto B;
        }
        else
        {
            cin.clear();
            fflush(stdin); // clear input buffer
            user_password();
        }
    }
    valid(IC);
}
}

```

// admin func

void covid_vaccination::add_vaccine_stock()

```

{
A:
    int option;
    fstream file;
    system("cls");
    title();
    cout << "\n\n\t\tAdd Vaccine To Center ";
    cout << "\n\n\t\t1. " << VacCenter;
    cout << "\n\n\t\t2. Back";
    cout << "\n\n\t\tEnter Option: ";
    cin >> option;
    switch (option)
    {
    case 1:
    {
        int x, add;
        file.open("vac_center.txt", ios::in);
        if (!file)
        {
            cout << "File not found" << endl;
            cout << "\n\n\n\t\tPress Any Key To Continue..";
            getch();
        }
    }
    }
}

```

```

        break;
    }
    file >> x;
    file.close();
    cout << "\n\n\t\tVaccine In Center: " << x;
    cout << "\n\n\t\tEnter Number Of Vaccines You Want To Add: ";
    while (!(cin >> add))
    {
        cout << "\n\n\t\tInvalid Input! Enter A Numeric Value: ";
        cin.clear();
        cin.ignore(numeric_limits<streamsize>::max(), '\n');
    }
    x = x + add;
    file.open("vac_center.txt", ios::out);
    file << x;
    file.close();
    cout << "\n\n\t\tVaccine In Center: " << x;
    cout << "\n\n\t\t" << add << " Vaccine Added";
    cout << "\n\n\n\t\tPress Any Key To Continue..";
    getch();
    break;
}
case 2:
    break;
default:
    retypeOpt();
    break;
}
}

void covid_vaccination::display_vaccine_stock()
{
    system("cls");
    vac_data data;
    ifstream ind;
    fstream file;
    int a;
    title();
    cout << "\n\n\t\t Available Vaccine Stock ";
    int i = 0;
    file.open("vac_center.txt");
    file >> a;
    ind.open("Vaccination_Records.bin", ios::binary | ios::in); // read data from the file
    ind.seekg(0, ios::beg); // Set the input file's position to the beginning of the file
    while (ind.read((char *)&data, sizeof(data))) // Read records from the input file until the end is
reached
    {
        i++; // Increment the record counter
    }
    ind.close();
    int s = a - i;
    file.close();
    cout << "\n\n\t\t " << VacCenter << " ";
    cout << "\n\n\t\tTotal Vaccine: " << a;
}

```

```

    cout << "\n\n\t\tAssigned Vaccine Person: " << i;
    cout << "\n\n\t\tAvailable Number Of Vaccines are: " << s;
    cout << "\n\n\t\tPress Any Key To Continue..";
    getch();
}

void covid_vaccination::vaccination_record()
{
    vac_data Data;
    system("cls");
    title();
    cout << "\n\n\t\t Assigned Vaccine Person \n\n\t\t";
    ifstream file("Vaccination_Records.bin", ios::binary);
    if (!file.is_open())
    {
        cerr << "Failed to open file";
    }
    // Get the size of the file
    file.seekg(0, ios::end);
    streamoff fileSize = file.tellg();
    file.seekg(0, ios::beg);

    // Allocate memory for the buffer
    char *buffer = new char[fileSize];

    // Read contents of file into buffer
    file.read(buffer, fileSize);

    // Calculate the number of data structures in the buffer
    int dataCount = fileSize / sizeof(Data);

    // Determine maximum length for each column
    int maxLength[5] = {0, 0, 0, 0, 0};
    for (int i = 0; i < dataCount; ++i)
    {
        vac_data d;
        memcpy(&d, &buffer[i * sizeof(Data)], sizeof(Data));
        maxLength[0] = max(maxLength[0], (int)string(d.uIC).length() + 1);
        maxLength[1] = max(maxLength[1], (int)string(d.uname).length() + 4);
        maxLength[2] = max(maxLength[2], 5);
        maxLength[3] = max(maxLength[3], 11);
        maxLength[4] = max(maxLength[4], 16);
    }

    // Print table header
    cout << left << setw(maxLength[0]) << "IC No."
        << left << setw(maxLength[1]) << "Name"
        << left << setw(maxLength[2]) << "Dose"
        << left << setw(maxLength[3]) << "Date"
        << left << setw(maxLength[4]) << "Manufacturer" << endl;

    // Print contents of data as a table
    for (int i = 0; i < dataCount; ++i)
    {

```

```

    vac_data d;
    memcpy(&d, &buffer[i * sizeof(Data)], sizeof(Data));
    cout << "\t\t";
    cout << left << setw(maxLength[0]) << d.uIC
        << left << setw(maxLength[1]) << d.uname
        << left << setw(maxLength[2]) << d.dose
        << left << setw(maxLength[3]) << d.udate
        << left << setw(maxLength[4]) << d.manufacturer << endl;
}
delete[] buffer;
cout << "\n\n\t\tPress Any Key To Continue..";
getch();
}

// user func
void covid_vaccination::apply_vaccine()
{
    vac_data data;
    vac_data datas;
    vac_data datac;
    fstream file;
    ifstream files;
    int length;
    int day, month, year;
    int result;
    char choice;
    int count = 0;

    system("cls");
    title();
    cout << "\n\n\t\t Apply Vaccine ";
A:
    cout << "\n\n\t\tEnter IC No.: ";
    fflush(stdin);
    cin >> data.uIC;
    if (strlen(data.uIC) != 12)
    {
        cout << "\n\t\tInvalid IC Number";
        char choice;
        cout << "\n\n\t\tEnter Y To Retype IC No. : ";
        cin >> choice;
        if (choice == 'y' || choice == 'Y')
        {
            goto A;
        }
        else
        {
            cin.clear();
            fflush(stdin); // clear input buffer
            goto D;
        }
    }
}
file.open("Vaccination_Records.bin", ios::in | ios::binary);
while (file.read((char *)&datac, sizeof(datac)))

```

```

{
    if (strcmp(data.uIC, datac.uIC) == 0)
    {
        count++;
    }
}
file.close();
files.open("Vaccination_Records.bin", ios::in | ios::binary);
if (!files)
{
    data.dose = 49;
    cout << "\n\t\tDose1 " << data.dose << ": ";
}
while (files.read((char *)&datas, sizeof(datas)))
{
    if (strcmp(data.uIC, datas.uIC) == 0) // check IC exist before or not
    {
        cout << "\n\t\tYou Have Already Apply Dose " << count;
        data.dose = (char)(count + 49);
        cout << ". You Are Applying";
        cout << "\n\t\tDose " << data.dose << ": ";
        strcpy(data.uname, datas.uname);
        cout << "\n\n\t\tName: " << data.uname << "\n";
        goto B;
    }
    else
    {
        data.dose = 49;
        cout << "\n\t\tDose " << data.dose << ": ";
        goto Z;
    }
}
Z:
files.close();
cout << "\n\t\tEnter Name: ";
cin >> data.uname;
B:
cout << "\n\t\tEnter date (DD/MM/YYYY): ";
cin >> data.udate;
result = sscanf(data.udate, "%d/%d/%d", &day, &month, &year);
if (result != 3)
{
    cout << "\t\tInvalid Date format.";
    cout << "\n\n\t\tEnter Y To Retype Date : ";
    cin >> choice;
    if (choice == 'y' || choice == 'Y')
    {
        goto B;
    }
    else
    {
        cin.clear();
        fflush(stdin); // clear input buffer
        goto D;
    }
}

```

```

    }
}
C:
cout << "\n\t\tEnter Vaccine Manufacture \n\t\t(Pfizer-BioNTech/AstraZeneca/Sinovac): ";
cin >> data.manufacturer;
length = strlen(data.manufacturer);
for (int i = 0; i < length; ++i)
{
    data.manufacturer[i] = tolower(data.manufacturer[i]);
}
if (strcmp(data.manufacturer, "pfizer-biontech") == 0)
{
    strcpy(data.manufacturer, "Pfizer-BioNTech");
}
else if (strcmp(data.manufacturer, "astrazeneca") == 0)
{
    strcpy(data.manufacturer, "AstraZeneca");
}
else if (strcmp(data.manufacturer, "sinovac") == 0)
{
    strcpy(data.manufacturer, "Sinovac");
}
else
{
    cout << "\n\t\tInvalid manufacturer";
    char choice;
    cout << "\n\n\t\tEnter Y To Retype Manufacturer : ";
    cin >> choice;
    if (choice == 'y' || choice == 'Y')
    {
        goto C;
    }
    else
    {
        cin.clear();
        fflush(stdin); // clear input buffer
        goto D;
    }
}
cout << "\n\n\t\tPlease Double-Check All Information \n\t\tFor Accuracy And Completeness Before Submitting.";
cout << "\n\n\t\tEnter Y To Confirm : ";
cin >> choice;
if (choice == 'y' || choice == 'Y')
{
    file.open("Vaccination_Records.bin", ios::app | ios::binary);
    file.write((char *)&data, sizeof(data));
    cout << "\n\n\t\tYou Have Made An Vaccination Record";
}
else
{
    cin.clear();
    fflush(stdin); // clear input buffer
    cout << "\n\n\t\tVaccination Record Canceled";
}

```

```

    }
D:
    file.close();
    cout << "\n\n\t\tPress Any Key To Continue.." << endl;
    getch();
}

void covid_vaccination::view_vaccination()
{
    vac_data datac;
    int count = 0;
    char bIC[13];
    char bname[100];
    fstream file;
    system("cls");
    title();
    cout << "\n\n\t\t Check Vaccination Record ";
    file.open("Vaccination_Records.bin", ios::binary | ios::in);
    if (!file) // check file open
    {
        cout << "\n\n\t\tThere Are No User Appointed";
        goto D;
    }
A:
    cout << "\n\n\t\tEnter IC No.: ";
    cin >> bIC;
    if (strlen(bIC) != 12)
    {
        cout << "\n\n\t\tInvalid IC Number";
        char choice;
        cout << "\n\n\t\tEnter Y To Retype IC No. : ";
        cin >> choice;
        if (choice == 'y' || choice == 'Y')
        {
            goto A;
        }
        else
        {
            cin.clear();
            fflush(stdin); // clear input buffer
            goto D;
        }
    }
    cout << "\n\n\t\tName: ";
    cin >> bname;
    while (file.read((char *)&datac, sizeof(datac)))
    {
        if (strcmp(bIC, datac.uIC) == 0 && strcmp(bname, datac.uname) == 0)
        {
            count++;
            {
                cout << endl
                    << "\t\tDose " << datac.dose << ": ";
            }
        }
    }
}

```

```

        cout.setf(ios::left, ios::adjustfield);
        cout.width(20);
        cout << endl
            << "\t\tDate: ";
        cout.setf(ios::right, ios::adjustfield);
        cout.width(25);
        cout << setfill(' ') << datac.update;
        ;

        cout.setf(ios::left, ios::adjustfield);
        cout.width(20);
        cout << endl
            << "\t\tManufacturer: ";
        cout.setf(ios::right, ios::adjustfield);
        cout.width(25);
        cout << setfill(' ') << datac.manufacturer;
    }
}
}
if (count == 0)
{
    cout << "\n\t\tRecord Not Found!";
}
file.close();
D:
cout << "\n\n\t\tPress Any Key To Continue..";
getch();
}

void covid_vaccination::delete_vaccination()
{
    vac_data datac;
    int count = 0;
    char bIC[13];
    char bname[100];
    fstream file;
    system("cls");
    title();
    cout << "\n\n\t\t Delete Vaccination Record ";
    file.open("Vaccination_Records.bin", ios::binary | ios::in);
    if (!file) // check file open
    {
        cout << "\n\t\tThere Are No User Appointed";
        goto D;
    }
    else
    {
        goto A;
    }
}
A:
cout << "\n\n\t\tEnter IC No.: ";
cin >> bIC;
if (strlen(bIC) != 12)
{

```



```

    cout << "\n\t\tInvalid IC Number";
    char choice;
    cout << "\n\n\t\tEnter Y To Retype IC No. : ";
    cin >> choice;
    if (choice == 'y' || choice == 'Y')
    {
        goto A;
    }
    else
    {
        cin.clear();
        fflush(stdin); // clear input buffer
        goto D;
    }
}
cout << "\n\t\tName: ";
cin >> bname;
while (file.read((char *)&datac, sizeof(datac)))
{
    if (strcmp(bIC, datac.uIC) == 0 && strcmp(bname, datac.uname) == 0)
    {
        count++;
        {
            cout << endl
                << "\t\tDose " << datac.dose << ": ";

            cout.setf(ios::left, ios::adjustfield);
            cout.width(20);
            cout << endl
                << "\t\tDate: ";
            cout.setf(ios::right, ios::adjustfield);
            cout.width(25);
            cout << setfill(' ') << datac.ude;
            ;

            cout.setf(ios::left, ios::adjustfield);
            cout.width(20);
            cout << endl
                << "\t\tManufacturer: ";
            cout.setf(ios::right, ios::adjustfield);
            cout.width(25);
            cout << setfill(' ') << datac.manufacturer;
        }
    }
}
file.close();
if (count == 0)
{
    cout << "\n\t\tRecord Not Found!";
    goto D;
}
cout << "\n\n\t\tEnter Y To Delete Vaccination Record : ";
char choice;
cin >> choice;

```

```

if (choice == 'y' || choice == 'Y')
{
    ifstream binaryFile("Vaccination_Records.bin", ios::binary);
    if (!binaryFile)
    {
        cerr << "Error Opening Vaccination Record File";
    }
    ofstream updatedBinaryFile("Vaccination_Records_temp.bin", ios::binary | ios::trunc);
    if (!updatedBinaryFile)
    {
        cerr << "Error Opening Vaccination Record Temp File";
        binaryFile.close();
    }
    while (binaryFile.read((char *)&datac, sizeof(datac)))
    {
        if (strcmp(datac.uIC, bIC) != 0)
        {
            updatedBinaryFile.write((const char *)&datac, sizeof(datac));
        }
    }
    binaryFile.close();
    updatedBinaryFile.close();
    remove("Vaccination_Records.bin");
    rename("Vaccination_Records_temp.bin", "Vaccination_Records.bin");
    cout << "\n\n\t\tYou Have Delete The Vaccination Record";
}
else
{
    cin.clear();
    fflush(stdin); // clear input buffer
    cout << "\n\n\t\tYou Have Cancel Delete Vaccination Record";
    goto D;
}
D:
cout << "\n\n\t\tPress Any Key To Continue..";
getch();
}

// both func

int main()
{
    covid_vaccination system;
    system.vac_center();
    system.menu();
}

```